

HIDE and Seek: Detecting Hallucinations in Language Models via Decoupled Representations

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Contemporary language models can generate fluent responses that are factually incorrect or unfaithful to the input context. We propose HIDE, a training-free, single-generation hallucination detector motivated by input–output representational dependence. HIDE applies an adapted HSIC-based score to selected hidden states collected during generation. We evaluate six open-source language models on four QA benchmarks covering faithfulness and factuality. In the original benchmark tables, HIDE has approximately 29% mean relative AUC improvement over the better of the evaluated perplexity and energy baselines within each model/dataset setting, and is competitive with multi-generation methods. A new matched comparison on 2,000 examples per dataset and model, using explicit first-answer-line stopping, finds that prompt attention mass outperforms HIDE in seven of eight settings. A selected-token-count control closely tracks HIDE, limiting a specifically nonlinear-dependence interpretation of its performance. Paired latency measurements extend to Gemma-2-27B and show model- and dataset-dependent overhead under standard HuggingFace inference. The historical approximately 51% reduction relative to the evaluated multi-pass methods is a baseline architectural comparison, not an optimized-serving measurement. These results identify both the practical scope and limitations of the proposed score.

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1. Introduction

Our evolving relationship with *truth* in the age of artificial intelligence is increasingly shaped by modern Language Models (LMs) (Brown et al. 2020; Chowdhery et al. 2023; Touvron et al. 2023). Despite their remarkable fluency, these models can generate content that is factually incorrect or unfaithful to the input context while remaining highly confident — a phenomenon commonly referred to as *hallucination* (Ji et al. 2023). Such hallucinated outputs, which often appear coherent and convincing, pose serious challenges to the reliable and trustworthy deployment of LMs, particularly as these systems are integrated into real-world applications at scale (Zhang et al. 2023; Weidinger et al. 2022; Huang et al. 2025; Maynez et al. 2020).

Hallucinations manifest in different forms. Huang et al. (2025) distinguish between *faithfulness hallucinations*, where the output contradicts the provided source or context, and *factuality hallucinations*, where the model generates information inconsistent with real-world facts. Prior work has attributed these failures to factors such as training data

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1 Code and retained experimental records: <https://github.com/C-anwoy/HIDE>.

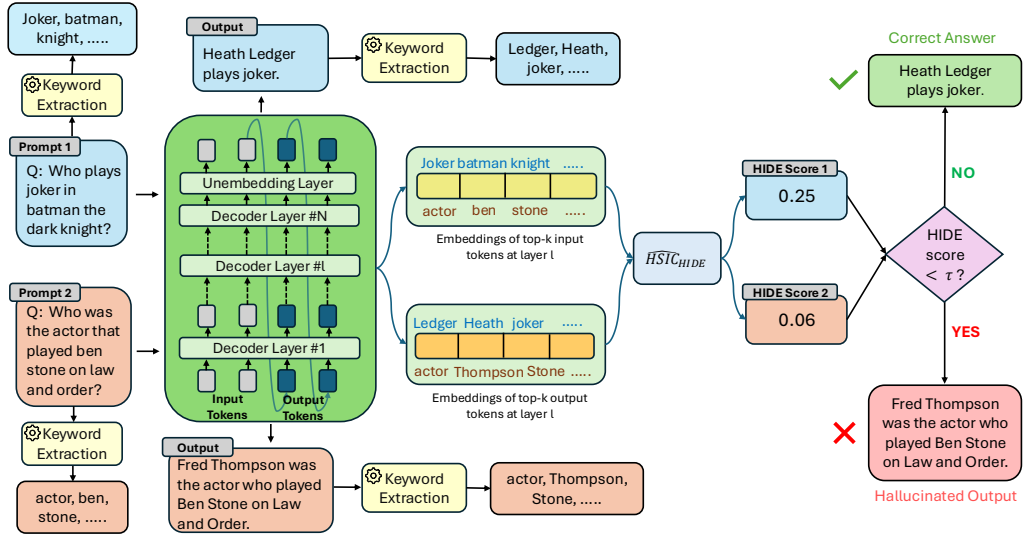


Figure 1: Our proposed method, HIDE, is a single-pass, training-free approach for hallucination detection. For a given input prompt, HIDE extracts representative tokens from both the input and the LM-generated output, computes a HIDE score using an adapted variant of unbiased HSIc on the corresponding hidden-state embeddings from a specific LM layer, and compares this score against a threshold to identify hallucinated generations.

artifacts, model architectures, and inference strategies (Lee et al. 2022; Onoe et al. 2022; Zheng, Huang, and Chang 2023). While the term “hallucination” itself is sometimes debated due to its anthropomorphic connotations (Maleki, Padmanabhan, and Dutta 2024), the underlying issue of unreliable generation remains a critical concern, motivating the development of effective hallucination detection methods.

A common approach to hallucination detection estimates model uncertainty using metrics such as perplexity (Ren et al. 2023), entropy (Malinin and Gales 2021), or energy-based scores (Liu et al. 2020), under the assumption that hallucinations correlate with low confidence. However, LMs are often confidently wrong, limiting the effectiveness of uncertainty-based signals (Simhi et al. 2025). Another line of work detects hallucinations via consistency checks, generating multiple outputs per input and comparing them using lexical or semantic similarity measures (Lin, Liu, and Shang 2022; Manakul, Liusie, and Gales 2023). While effective, these approaches incur substantial computational overhead due to repeated auto-regressive decoding. Recent mechanistic studies further show that hallucination-related cues are encoded within LM hidden states (Azaria and Mitchell 2023; Ji et al. 2024; Duan, Yang, and Tam 2024), and some methods train probing classifiers on these representations. In contrast, our goal is to directly leverage internal representations for hallucination detection *without* supervised training and *without* requiring multiple generations.

To this end, we propose **HIDE** (Hallucination DetectIOn via Decoupled REpresentations), a training-free method that detects hallucinations by measuring the statistical dependence between an LM’s internal representations of the input and the generated output. As illustrated in Figure 1, HIDE extracts hidden states corresponding to key tokens from both input and output sequences and applies an adapted variant

of the Hilbert–Schmidt Independence Criterion (HSIC) to quantify representational coupling. Low dependence indicates representational decoupling, signaling a potential hallucination. Unlike multi-pass methods, HIDE requires only a single auto-regressive generation per input, avoiding additional sampled responses; the resulting latency depends on the inference implementation.

We evaluate HIDE on four question answering benchmarks — SQuAD (Rajpurkar et al. 2016) and RACE (Lai et al. 2017) for faithfulness hallucinations, and Natural Questions (Kwiatkowski et al. 2019) and TriviaQA (Joshi et al. 2017) for factuality hallucinations — across six open-source LMs of varying scales and training paradigms, including base and instruction-tuned variants of Llama-3.2-3B, Llama-3-8B (Grattafiori et al. 2024), and Gemma-2-9B (Team et al. 2024). These QA settings allow objective verification of hallucinations against reference answers; we additionally include dialogue experiments to assess generalization beyond QA.

Our contributions and empirical scope are:

- We introduce a training-free, single-generation HSIC-based detector and evaluate it on faithfulness and closed-book factuality benchmarks.
- We report the original benchmark comparisons separately from a new matched attention/norm comparison. In the corrected comparison, attention is stronger in seven of eight settings, and a selected-token-count control challenges a specifically nonlinear interpretation of the HIDE score.
- Despite requiring only one generation, HIDE can be compared with multi-generation alternatives in the original benchmark. The historical approximately 51% timing reduction is restricted to standard HuggingFace inference. Additional paired measurements extend to Gemma-2-27B without claiming constant overhead or optimized-serving speedups.

The historical ablations show limited sensitivity to several layer and kernel choices; the selected-token-count diagnostic qualifies the interpretation of this stability (§7). Together, these experiments characterize a single-generation detection score while identifying substantial limits on its mechanistic interpretation and advantage over simpler controls.

2. Background and Related Work

Though hallucination is often deemed to be essential for the creativity of LMs (Lee 2023; Chakraborty and Masud 2024), it remains one of the most frequently criticized shortcomings of these models. While some studies argue that hallucinations are an inherent consequence of their stochastic nature, often known as the ‘stochastic parrots’ phenomenon (Bender et al. 2021), many others attribute the issue primarily to artifacts in the data that these models ingest. There is a substantial body of research addressing diverse aspects of hallucination in LMs – we briefly summarize a few of these works related to two major dimensions: sources of hallucination and hallucination detection.

2.1 Sources of Hallucination

Hallucinations in LMs have been attributed to a complex interplay of factors spanning data quality, model architecture, and inference processes. On the data front, pre-training on vast and often imperfect corpora introduces risks of propagating misinformation, factual inaccuracies, and social biases (Lin, Hilton, and Evans 2022; Lee et al. 2022; Hernandez et al. 2022; Paullada et al. 2021; Narayanan Venkit et al. 2023; Ladhak et al. 2023; Kasai et al. 2023). The ever-increasing scale of training data, while enabling greater

generalization, can also amplify errors and result in LMs memorizing, rather than reasoning about information.

At the level of model training and design, certain architectural limitations – such as insufficient bidirectional representation or attention mechanism failures – can introduce systematic vulnerabilities to hallucination (Li et al. 2023; Liu et al. 2023; Wang and Sennrich 2020; Sharma et al. 2024). Moreover, the alignment process that tunes LMs to follow human preferences may occasionally introduce a mismatch between model capabilities and desired behaviors, leading to unexpected model outputs.

Inference strategies, especially those introducing stochasticity or prioritizing diversity (such as top- k or nucleus sampling), have also been linked to hallucination (Fan, Lewis, and Dauphin 2018; Holtzman et al. 2020; Dziri et al. 2021; Chuang et al. 2024). Additional challenges arise from limitations in context integration and representational bottlenecks in output layers (Liu et al. 2024; Shi et al. 2024; Yang et al. 2018; Chang and McCallum 2022). Together, these findings indicate that hallucinations arise not from a single source but from interactions among data, architecture, and inference.

2.2 Hallucination Detection

Given the broad impact of hallucination, a wide range of detection strategies has been proposed. Many early efforts focused on evaluating model confidence, with the intuition that LMs are less certain when hallucinating. Metrics such as perplexity, entropy, and energy-based scores (Ren et al. 2023; Malinin and Gales 2021; Liu et al. 2020) have been explored for this purpose. Another family of approaches leverages output consistency: by sampling multiple generations for the same prompt, one can measure lexical or semantic agreement across responses (Lin, Hilton, and Evans 2022; Durmus, He, and Diab 2020; Scialom et al. 2021). High variability is often taken as a proxy for hallucination, and techniques such as BERTScore, NLI-based classification, and question-answering-based evaluation have all been applied (Falke et al. 2019; Maynez et al. 2020; Nan et al. 2021; Chen et al. 2024b). However, these multi-pass strategies are computationally demanding and poorly suited to low-latency settings, as they require generating and analyzing numerous outputs for each input.

Other detection frameworks rely on verifying factuality by grounding model outputs in external knowledge sources (Min et al. 2023; Huo, Arabzadeh, and Clarke 2023; Chern et al. 2023; Chen et al. 2024b). While these methods can be effective, they require access to curated external databases and ongoing maintenance, and may not generalize across domains.

More recently, studies have explored the use of LMs themselves as evaluators – either via prompt engineering that elicits self-critique, or by prompting additional LMs to judge the factuality or faithfulness of generated content (Xiong et al. 2024; Kadavath et al. 2022; Manakul, Liusie, and Gales 2023; Chiang and Lee 2023; Luo, Xie, and Ananiadou 2023; Laban et al. 2023; Miao, Teh, and Rainforth 2023). These meta-evaluation techniques provide flexibility in black-box settings, but can inherit the blind spots of the underlying models and are themselves subject to hallucination.

Mechanistic Interpretability and Representation-based Hallucination Detection. A growing body of work in mechanistic interpretability investigates how hallucinations are reflected in the internal representations of language models. Several studies demonstrate that hidden states encode signals predictive of factuality or truthfulness, and leverage supervised probes to extract these signals for hallucination detection or analysis (Azaria and Mitchell 2023; Ji et al. 2024; Duan, Yang, and Tam 2024; Obeso et al. 2025). Recent

work has also explored latent-space diversity, activation geometry, and representation dynamics as indicators of hallucinated generation (Chen et al. 2024a; Geva et al. 2023). Collectively, these studies provide strong evidence that hallucinations are systematically encoded within LM hidden states. While effective, most representation-based detection methods rely on supervised training of task- and model-specific classifiers, or require dataset-dependent calibration. As a result, their performance can degrade under domain or task shifts, and their deployment typically requires access to labeled hallucination data and retraining for new settings. In contrast, HIDE does not train any auxiliary model and instead directly measures statistical dependence between input and output representations at inference time. This design makes HIDE training-free and task-agnostic, enabling deployment without labeled hallucination data. In Section 6, we empirically compare HIDE against a strong supervised probing baseline and show that, while probes perform well on their training domain, HIDE exhibits more robust generalization to unseen datasets without any training.

2.3 Background of Hallucination Detection Measures

Here we briefly discuss the mathematical formulation of a few hallucination detection measures, which are used as baselines in this work. Since our proposed method doesn't involve any training, we compare only against other training-free methods.

Assume that $\mathbf{S}_{in} = \{s_1, s_2, \dots, s_I\}$ is the input prompt, and $\mathbf{S}_{out} = \{t_1, t_2, \dots, t_O\}$ is the corresponding output generated by the LM, parameterized by θ , where $\mathbf{S}_{out}$ is a sequence of O tokens. In case of measures requiring multiple generations for the same input $\mathbf{S}_{in}$, assume $\mathcal{S} = [\mathbf{S}_{out}^1, \mathbf{S}_{out}^2, \dots, \mathbf{S}_{out}^N]$ to be the collection of N generations, where diversity is ensured through stochastic sampling strategies, like top- k , nucleus or temperature sampling. Also, $\mathbb{P}_{LM}(\cdot)$ denotes the probability distribution generated by the LM for any position in the sequence.

Perplexity. Perplexity, as a token-level uncertainty measure, can be utilized for evaluating potential hallucinations (Ren et al. 2023). Perplexity is monotonically related to the mean negative log-likelihood (MNLL) of the output sequence, and hence MNLL can be utilized as a tractable proxy for perplexity as a hallucination detection measure (Chen et al. 2024a). Formally, MNLL can be defined as: $\text{MNLL}(\mathbf{S}_{out}|\mathbf{S}_{in};\theta) = -\frac{1}{O} \sum_{i=1}^O \log \mathbb{P}_{LM}(t_i|t_{<i}, \mathbf{S}_{in})$. A lower value of this measure indicates that the model is more confident about its generation.

Energy Score. The Energy Score (Liu et al. 2020) approaches hallucination detection from a different perspective by leveraging energy-based models. Unlike softmax-based confidence scores, energy scores are aligned with the probability density of the inputs. Let $\mathbb{L}_{LM}(\cdot)$ denote the logit distribution over the set of tokens in the vocabulary $\mathcal{V}$, generated by the LM, such that $\mathbb{P}_{LM}(\cdot) = \text{softmax}(\mathbb{L}_{LM}(\cdot))$. The energy score, for a temperature parameter T , can then be defined as: $\text{Energy}(\mathbf{S}_{in};\theta) = -T \cdot \log \sum_{t \in \mathcal{V}} e^{\mathbb{L}_{LM}(t|\mathbf{S}_{in})/T}$. The energy score is generally used for out-of-distribution detection, where samples with higher energies can be interpreted as datapoints with a lower likelihood of occurrence.

Length-Normalized Entropy. Unlike perplexity which utilizes a single generation for quantifying sequence-level uncertainty, length-normalized entropy (LN-Entropy) uses multiple generations to measure the degree of hallucination (Malinin and Gales 2021). LN-Entropy is defined as: $\text{LNE}(\mathcal{S}|\mathbf{S}_{in};\theta) = -\mathbb{E}_{\mathbf{S}_{out} \in \mathcal{S}} \left[\frac{1}{|\mathbf{S}_{out}|} \sum_{i=1}^{|\mathbf{S}_{out}|} \log \mathbb{P}_{LM}(t_i|t_{<i}, \mathbf{S}_{in}) \right]$. It

Method	Uses	Checks	No. of Generations
	Uncertainty	Consistency	Required
Perplexity	✓	✗	1
Energy	✓	✗	1
Length-Normalized Entropy	✓	✗	N
Lexical Similarity	✗	✓	N
Eigenscore	✗	✓	N
HIDE (Ours)	✗	✗	1

Table 1: Comparison of some popular training-free hallucination detection techniques. Most methods rely on either model uncertainty or consistency across multiple generations (N) – approaches relying on the latter show increased latency and compute cost. HIDE, on the other hand, requires only a single generation and leverages internal representation independence, offering a lightweight yet effective alternative.

is assumed that when an LM generates hallucinated content, it is typically uncertain, resulting in an output distribution with higher entropy (Kadavath et al. 2022).

Lexical Similarity. Lexical Similarity (LS) (Lin, Liu, and Shang 2022) aims to quantify the similarities between the multiple generations from an LM corresponding to the same input prompt, with the assumption that LS will be low in case of hallucinations. It can formally defined as: $LS(\mathcal{S}|\mathbf{S}_{in}; \theta) = \frac{2}{N(N-1)} \sum_{i=1}^N \sum_{j=i+1}^N sim(\mathbf{S}_{out}^i, \mathbf{S}_{out}^j)$, where, $sim(\cdot, \cdot)$ can be any function to quantify similarity, like Rouge-L (Lin 2004).

EigenScore. EigenScore (Chen et al. 2024a) leverages the hidden states of LMs for hallucination detection. It measures semantic diversity in the latent space of LMs’ internal representations by calculating the log-determinant of the covariance matrix of multiple output representations: $ES(\mathcal{S}|\mathbf{S}_{in}; \theta) = \frac{1}{N} \log \det(\Sigma + \alpha \cdot \mathbb{I}_N) = \frac{1}{N} \sum_{i=1}^N \log(\lambda_i)$, where, the covariance matrix of the output representations is given by: $\Sigma = Z^T \cdot C_d \cdot Z$ with $Z = [z_1, z_2, \dots, z_N] \in \mathbb{R}^{d \times N}$ being the embedding matrix of the N generated outputs and $C_d = \mathbb{I}_d - \frac{1}{d} \mathbf{1}_d \mathbf{1}_d^T$ being the centering matrix. The set of eigenvalues of the regularized covariance matrix $\Sigma + \alpha \cdot \mathbb{I}$ is given by $\lambda = \{\lambda_1, \lambda_2, \dots, \lambda_N\}$. It is assumed that when the LM is not hallucinating, the representations of the outputs will be highly correlated, thereby having $\lambda_i = 0$ for most i ’s.

Table 1 presents a summary of the characteristics of the discussed hallucination detection measures, and compares our proposed method with these techniques.

3. Theoretical Foundations

We propose that linguistic hallucination in LMs often stems from a statistical decoupling between the model’s internal representation of the input and that of the generated output. A natural question that arises is whether simpler geometric similarity measures, such as cosine similarity between aggregated input and output representations, are sufficient to capture representational coupling. However, the relationship between faithful input-output representations in LLMs is often highly non-linear, shaped by attention, residual connections, and layer-wise transformations. As a result, linear similarity measures may fail to detect meaningful dependence even when representations are strongly coupled in a non-linear sense (c.f. §7.1). To address this, we adopt the Hilbert–Schmidt Independence Criterion (HSIC), a kernel-based, non-parametric measure of statistical

dependence. HSIC is sensitive to arbitrary non-linear relationships between random variables and evaluates dependence in a reproducing kernel Hilbert space, making it well-suited for capturing complex interactions between input and output representations in deep neural networks.

We investigate whether lower values of the proposed score are associated with incorrect generation, without treating this association as evidence that hallucination is caused by statistical independence (§§4 and 6). To quantify this phenomenon, we leverage the Hilbert-Schmidt Independence Criterion (HSIC), a non-parametric measure that can capture complex, non-linear relationships in high-dimensional spaces, such as LMs’ hidden states. We first introduce HSIC and discuss its properties, motivating its adaptation to our method for hallucination detection.

3.1 Measuring Representational Dependence

We hypothesize that hallucinations arise when an LM’s internal representations of the input and generated output become statistically decoupled. While faithful generations preserve a strong dependence between these representations, hallucinated outputs exhibit weakened or near-independent representations. Kernel methods can represent nonlinear relationships that simple geometric similarities may miss. Their potential expressiveness does not by itself establish that the configured finite-sample score uses this information in practice. To this end, we adopt the Hilbert-Schmidt Independence Criterion (HSIC), a kernel-based, non-parametric measure that can detect arbitrary statistical dependence. We provide the formal definition of HSIC, its independence guarantees, and relevant theoretical properties in Appendix A.1.

3.2 Empirical Estimation

In practice, we estimate HSIC using an unbiased estimator that remains reliable under small sample sizes, which is critical for hallucination detection where only a limited number of input and output tokens are available. The estimator operates on kernelized hidden-state representations extracted from a single forward pass of the LM. Details of the estimator, including its derivation and statistical properties, are provided in Appendix A.2.

4. HIDE: Hallucination Detection via Decoupled Representations

Building upon the theoretical foundations of HSIC, as introduced in §3, we propose HIDE, a single-pass approach for detecting hallucinations in LMs by investigating the internal model representations for input and output. HIDE is based on the premise that hallucinations manifest as a quantifiable statistical decoupling between the LM’s internal representations of the input sequence and the generated output. We hypothesize that:

- (i) **Sound Generation:** When an LM generates content that is grounded in the input context or in its parametric knowledge, its internal representation of the output, $H_{out}^{(\ell)}$, remains strongly statistically dependent on the internal representation of the input, $H_{in}^{(\ell)}$, at a given layer ℓ . This strong coupling should result in a substantially positive HSIC value.

- (ii) **Hallucinated Generation:** Conversely, when an LM hallucinates, the generative process decouples the output representation $H_{out}^{(\ell)}$ from the input representation $H_{in}^{(\ell)}$. As the generated content diverges, these representations become increasingly independent. In such cases, their joint distribution approaches the product of their marginals, leading to an HSIC value close to zero, i.e., $\text{HSIC}(H_{in}^{(\ell)}, H_{out}^{(\ell)}) \approx 0$.

HIDE leverages this hypothesized drop in HSIC to detect hallucinations. A sufficiently low empirical HSIC score between selected input and output representations suggests a potential hallucination.

4.1 Problem Formulation

As defined previously in §2.3, $\mathbf{S}_{in} = \{s_1, s_2, \dots, s_I\}$ is an input token sequence of length I , and $\mathbf{S}_{out} = \{t_1, t_2, \dots, t_O\}$ is the corresponding LM-generated output sequence of length O . For a chosen layer ℓ , the matrix of hidden states for the input sequence is $\mathbf{H}_{in}^{(\ell)} \in \mathbb{R}^{I \times d}$, where $\mathbf{h}_{i,in}^{(\ell)} \in \mathbb{R}^d$ is the hidden state for input token s_i , and d is the hidden state dimensionality. Similarly, for the output sequence, $\mathbf{H}_{out}^{(\ell)} \in \mathbb{R}^{O \times d}$, with rows $\mathbf{h}_{j,out}^{(\ell)} \in \mathbb{R}^d$.

To apply HSIC, we define random variables X and Y based on these representations:

Definition 1

Let $\mathcal{T}_{in}$ be the set of all possible token types that can be selected from an input sequence, and $\mathcal{T}_{out}$ be the set of all possible token types from an output sequence. We define two random variables, X and Y , as :

- $X : \mathcal{T}_{in} \rightarrow \mathbb{R}^d$, where for any selected input token type $\sigma_{in} \in \mathcal{T}_{in}$ (which, in a specific instance, corresponds to a token $s_i \in \mathbf{S}_{in}$), $X(\sigma_{in}) = \mathbf{h}_{i,in}^{(\ell)}$. The codomain of X is $\mathcal{X} = \mathbb{R}^d$.
- $Y : \mathcal{T}_{out} \rightarrow \mathbb{R}^d$, where for any selected output token type $\sigma_{out} \in \mathcal{T}_{out}$ (which, in a specific instance, corresponds to a token $t_j \in \mathbf{S}_{out}$), $Y(\sigma_{out}) = \mathbf{h}_{j,out}^{(\ell)}$. The codomain of Y is $\mathcal{Y} = \mathbb{R}^d$.

For a single input-output pair $(\mathbf{S}_{in}, \mathbf{S}_{out})$, we independently select n_{eff} tokens from $\mathbf{S}_{in}$ and $\mathbf{S}_{out}$, respectively, to obtain samples $\{x_1, \dots, x_{n_{eff}}\}$ for X and $\{y_1, \dots, y_{n_{eff}}\}$ for Y . These form the two sets of observations for calculating empirical HSIC.

4.2 HIDE Score Computation

The HIDE framework for detecting hallucination in a single generation pass is summarized in Algorithm 1. By *single-pass*, we mean that HIDE requires no additional forward passes of the language model beyond the standard auto-regressive generation of a single output sequence. The keyword extraction step operates on the generated text and uses a lightweight embedding-based ranking model; it does not alter the LM's generation process or require additional decoding steps. Consequently, the dominant computational cost of HIDE remains the original generation pass, distinguishing it from multi-pass detection methods that require multiple generations per input.

A key component of the proposed method is the computation of our HSIC-based score. The standard unbiased HSIC estimator, $\widehat{\text{HSIC}}_u$ (c.f. Equation A.3), provides

desirable statistical properties but is formally defined for $n \geq 4$ samples. In practical scenarios, particularly when processing short input or output sequences, the effective number of selected tokens, n_{eff} , may fall below this threshold. To ensure the applicability of our framework across all sequence lengths and maintain numerical stability, we employ a pragmatically adapted HSIC-like score, denoted by $\widehat{\text{HSIC}}_{\text{HIDE}}$. This adaptation modifies the denominators in the standard unbiased HSIC estimator (Equation A.3) by replacing terms of the form $(n - c)$ with n , and substituting the overall scaling factor $1/(n(n - 3))$ with $1/n^2$.

Definition 2

For an effective sample size $n \geq 1$, the $\widehat{\text{HSIC}}_{\text{HIDE}}$ score is computed as:

$$\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) = \frac{1}{n^2} \left[\text{Tr}(\tilde{\mathbf{K}}_X \tilde{\mathbf{K}}_Y) + \frac{(\mathbf{1}^T \tilde{\mathbf{K}}_X \mathbf{1})(\mathbf{1}^T \tilde{\mathbf{K}}_Y \mathbf{1})}{n^2} - \frac{2}{n} \mathbf{1}^T \tilde{\mathbf{K}}_X \tilde{\mathbf{K}}_Y \mathbf{1} \right] \quad (1)$$

where $\tilde{\mathbf{K}}_X, \tilde{\mathbf{K}}_Y$ are the $n \times n$ Gram matrices with zeroed diagonals, and $\mathbf{1}$ is an $n \times 1$ vector of ones.

While this adaptation introduces bias relative to the standard $\widehat{\text{HSIC}}_u$, it yields a score that is well-defined and numerically stable for any $n \geq 1$. This practical modification allows for consistent application of our method across all encountered data instances. We posit that $\widehat{\text{HSIC}}_{\text{HIDE}}$ still serves as a valuable and usable heuristic for quantifying the relative strength of dependence, demonstrating its empirical effectiveness in detecting hallucinations in §6.

Theoretical Properties. The HIDE score inherits several desirable properties from HSIC, including sensitivity to non-linear dependence and robustness under representation transformations. A formal analysis of these properties is provided in Appendix A.3.

4.3 Methodological Choices

Our design involves two key choices that must be fixed before the HIDE score can be evaluated: (i) how to align the *token sets* drawn from the input-output sequences so that the *same* number of samples are drawn for both random variables X and Y , and (ii) from which *decoder layer*, $\ell \in \{1, \dots, L\}$ the hidden-state matrices $\mathbf{H}_{\text{in}}^{(\ell)}$ and $\mathbf{H}_{\text{out}}^{(\ell)}$ are taken. We formalise the two choices below and motivate the instantiations that are used throughout the paper.

4.3.1 Strategy for Aligning the Sets of Input and Output Samples. Let $\mathbf{H}_{\text{in}}^{(\ell)} \in \mathbb{R}^{I \times d}$ and $\mathbf{H}_{\text{out}}^{(\ell)} \in \mathbb{R}^{O \times d}$ be the matrices of hidden states for the input and output sequences, respectively, at layer ℓ , where I and O may differ by in magnitude. To guarantee that X and Y are constructed from the *same* effective sample size, n_{eff} , we consider two principled strategies:

- (i) **Keyword-based Subsampling.** Let n_{eff} be our *token budget*, i.e., the number of representative tokens we want to select from both the input and output. Since, in some cases, the input or the output can be smaller than our token budget, we set the value of n_{eff} to be the minimum of our chosen token budget, input length and output length. We then extract a set of n_{eff} salient tokens $\mathcal{K}_{\text{in}} \subseteq \mathbf{S}_{\text{in}}$ and $\mathcal{K}_{\text{out}} \subseteq \mathbf{S}_{\text{out}}$ using KeyBERT,

Algorithm 1 HIDE: Hallucination Detection via Decoupled Representations

Input: Input sequence $\mathbf{S}_{in}$, Language model M , Layer index ℓ , Token budget n_{eff} , RBF kernel bandwidth γ , Threshold τ

Output: Decision: Hallucination or Non-hallucination

```

1: procedure DETECTHALLUCINATION( $\mathbf{S}_{in}, M, \ell, n_{\text{eff}}, \gamma, \tau$ )
2:   Initialize storage:
    $\mathbf{H}_{in}^{(\ell)} \leftarrow []$ ,  $\mathbf{H}_{out}^{(\ell)} \leftarrow []$ 
3:   Feed input  $\mathbf{S}_{in}$  and auto-regressively generate output  $\mathbf{S}_{out}$ :
4:   for each token  $t$  generated by  $M$  during inference do
5:     Compute hidden states at layer  $\ell$ 
6:     if  $t \in \mathbf{S}_{in}$  then
7:       Append its hidden representation to  $\mathbf{H}_{in}^{(\ell)}$ 
8:     else
9:       Append to  $\mathbf{H}_{out}^{(\ell)}$ 
10:    end if
11:  end for
12:  Now  $\mathbf{H}_{in}^{(\ell)}, \mathbf{H}_{out}^{(\ell)}$  hold all relevant states
13:   $n_{\text{eff}} \leftarrow \min(n_{\text{eff}}, |\mathbf{S}_{in}|, |\mathbf{S}_{out}|)$ 
14:  if  $n_{\text{eff}} < 1$  then
15:    return Undetermined
16:  end if
17:  Identify top- $n_{\text{eff}}$  key-tokens:
   $\mathcal{K}_{in} \leftarrow \text{KeyBERT}(\mathbf{S}_{in}, n_{\text{eff}})$ 
   $\mathcal{K}_{out} \leftarrow \text{KeyBERT}(\mathbf{S}_{out}, n_{\text{eff}})$ 
18:  Collect hidden vectors for selected tokens:
   $\mathbf{X} \leftarrow \{\mathbf{h}_{i,in}^{(\ell)} \mid \text{token}_i \in \mathcal{K}_{in}\}$ 
   $\mathbf{Y} \leftarrow \{\mathbf{h}_{j,out}^{(\ell)} \mid \text{token}_j \in \mathcal{K}_{out}\}$ 
19:  Build RBF kernel  $k_{\text{rbf}}$  with bandwidth  $\gamma$ 
20:  Compute HIDE score:
   $\text{score} \leftarrow \widehat{\text{HSIC}}_{\text{HIDE}}(\mathbf{X}, \mathbf{Y}; k_{\text{rbf}})$ 
21:  if  $\text{score} < \tau$  then
22:    return Hallucination
23:  else
24:    return Non-hallucination
25:  end if
26: end procedure

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which acts as a maximal-marginal-relevance ranker. The aligned sample sets are then $\mathcal{X} = \{\mathbf{h}_{i,in}^{(\ell)} \mid s_i \in \mathcal{K}_{in}^{(n_{\text{eff}})}\}$ and $\mathcal{Y} = \{\mathbf{h}_{j,out}^{(\ell)} \mid t_j \in \mathcal{K}_{out}^{(n_{\text{eff}})}\}$, where the superscript n_{eff} denotes truncation to the top- n_{eff} representative tokens. In practice we use $n_{\text{eff}} = 20$ for our experiments, unless explicitly specified otherwise. We discuss more on the effect of varying our token budget in §7.2.

- (ii) **Rank-truncated SVD Alignment.** Alternatively, we avoid lexical heuristics altogether and enforce equal sample size by projecting *both* matrices $\mathbf{H}_{in}^{(\ell)}$ and $\mathbf{H}_{out}^{(\ell)}$ to subspaces with the same rank (i.e., n_{eff}). Concretely, for $\mathbf{H}_{in}^{(\ell)}$ we compute a thin SVD

$\mathbf{H}_{\text{in}}^{(\ell)} = \mathbf{U}_X \Sigma_X \mathbf{V}_X^\top$ and retain the n_{eff} largest singular directions,

$$\mathbf{X}_r = \Sigma_X^{(n_{\text{eff}})} (\mathbf{V}_X^{(n_{\text{eff}})})^\top \in \mathbb{R}^{n_{\text{eff}} \times d}. \quad (2)$$

An analogous projection $\mathbf{Y}_r$ is obtained for $\mathbf{H}_{\text{out}}^{(\ell)}$, yielding aligned sample sets $\mathcal{X} = \{\mathbf{x}_r^{(1)}, \dots, \mathbf{x}_r^{(n_{\text{eff}})}\}$ and $\mathcal{Y} = \{\mathbf{y}_r^{(1)}, \dots, \mathbf{y}_r^{(n_{\text{eff}})}\}$. As both $\mathbf{X}_r$ and $\mathbf{Y}_r$ now contain exactly n_{eff} *orthogonal* rows, the empirical Gram matrices in Equation (1) have identical dimensions, eliminating the need for any further padding or truncation. We discuss the performances of HIDE with both these alignment strategies in §7.6. Unless stated otherwise, we adopt the keyword-based subsampling strategy for our experiments.

4.3.2 Layer Selection. Each decoder layer captures different levels of abstraction; therefore, the choice of ℓ can, in principle, influence the measured dependence. Based on prior work suggesting that representations corresponding to the middle layers are often richer (Skean et al. 2025) and provide stronger sentence-level semantic information suitable for detecting inconsistencies (Azaria and Mitchell 2023), we fix $\ell = \ell_{\text{mid}}$ for all our experiments to avoid an additional hyperparameter sweep. However, our framework is flexible to any layer choice and we observe the performance of HIDE to be almost agnostic to the choice of ℓ (§7.3).

4.4 Structural Analysis of Decoupling

We examine associations between the HIDE score and attention-mediated information flow. These are observational diagnostics, not evidence that statistical decoupling causes hallucination. In particular, attention to a question in closed-book QA need not imply factual correctness, while attention to supplied evidence can support faithful generation.

At the selected layer ℓ , let $\alpha_{j,i}^{(h)}$ denote the causal self-attention weight from an observed generated-token position j to prompt token i in head h . Let J be the number of generated-token positions with available cached states. We use

$$\bar{\alpha}_{j,i} = H^{-1} \sum_{h=1}^H \alpha_{j,i}^{(h)}, \quad (3)$$

$$\Omega = J^{-1} \sum_{j=1}^J \sum_{i=1}^I \bar{\alpha}_{j,i}, \quad (4)$$

$$\tilde{\Delta}_{\text{in}} = J^{-1} \sum_{j=1}^J \left\| \sum_{i=1}^I \bar{\alpha}_{j,i} h_{i,\text{in}}^{(\ell-1)} \right\|_2. \quad (5)$$

The norm is an *unprojected input-state proxy*. It omits the learned value and output projections and therefore is not the actual input-derived residual update. Attention and preceding-layer states are aligned to the layer producing the hidden states used by HIDE. Both proxies average over all available generated-token positions and use no keyword extraction. Cached generation excludes the final unforwarded token; if no output state is available, all three scores use the recorded zero fallback. The standalone

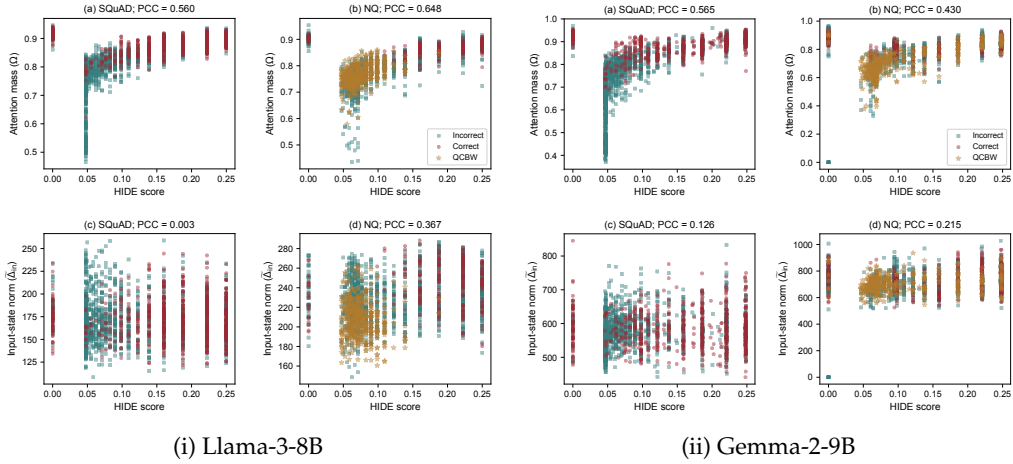


Figure 2: Observational information-flow diagnostics on the corrected cohorts: 2,000 examples per model/dataset, seed 42, greedy first-answer-line stopping. Columns show SQuAD and NQ; rows show prompt attention mass Ω and the unprojected input-state norm $\tilde{\Delta}_{in}$. Correctness uses answer-reference sentence similarity > 0.9 . Teal squares denote incorrect answers, red circles correct answers, and gold stars the NQ QCBW subset (upper question-answer-similarity quartile among incorrect answers, including cutoff ties). Every example is included; PCC is computed before display rounding, without score filtering or class balancing. These associations do not establish causality.

detection comparison uses the raw positive direction of each score without choosing its sign on test labels.

Figure 2 is recomputed on the corrected, matched cohorts described in Section 5: 2,000 examples per model/dataset, with no score filtering or class rebalancing. For Llama-3-8B, $PCC(HIDE, \Omega)$ is 0.560 on SQuAD and 0.648 on NQ; $PCC(HIDE, \tilde{\Delta}_{in})$ is 0.003 and 0.367, respectively. For Gemma-2-9B, the corresponding attention correlations are 0.565 and 0.430, and the norm correlations are explicitly *positive*: 0.126 on SQuAD and 0.215 on NQ. These values replace the earlier figure’s correlations, which used a different generation/selection protocol. Thus the previously foregrounded inverse “magnitude versus precision” relationship is not a general result. Neither update magnitude nor its correlation with HIDE establishes faithful information transfer or a causal mechanism.

Question-conditioned but wrong answers. For NQ, star markers identify incorrect answers in the upper quartile of question-answer embedding similarity among incorrect answers (including ties at the cutoff). This QCBW subset controls descriptively for superficial topicality; it is not an intervention and does not identify causal circuits. All other examples remain in the figure and in the reported correlations. The standalone baseline results in Section 6.2 provide a direct test of whether HIDE adds predictive value beyond these proxies.

5. Experimental Setup

Evaluation Datasets. Following the setup of Hong et al. (2024), we evaluate HIDE on four benchmark QA-datasets, as discussed below, to assess its effectiveness in detecting both faithfulness and factuality hallucinations:

- (i) **RACE (ReAding Comprehension dataset from Examinations)** (Lai et al. 2017) is a multiple-choice reading comprehension dataset from English exams for Chinese students, featuring 27,933 passages and 97,867 questions. We use the test split of the ‘high’ subset containing 3,498 questions. RACE is challenging due to its focus on inference and reasoning. RACE is used to evaluate *faithfulness hallucinations*, as answers must be supported by explicit information in the provided passage, making inference and grounding critical.
- (ii) **SQuAD (Stanford Question Answering Dataset)** (Rajpurkar et al. 2016) is a widely used benchmark for extractive question answering, containing crowd-sourced questions on Wikipedia articles. SQuAD-v1.1 contains only questions for which answers are present within the provided context, while SQuAD-v2.0 extends it to include both answerable and unanswerable questions. Following Chen et al. (2024a), we use only answerable questions from the development split of v2.0, retaining 5,928 context-question-answer pairs. SQuAD also targets *faithfulness hallucinations*, as the correct answer must be a text span from the passage, requiring the model to remain faithful to the context.
- (iii) **NQ (Natural Questions)** (Kwiatkowski et al. 2019) is a large-scale question answering dataset consisting of real, anonymized queries issued to the Google search engine. Following Lin, Trivedi, and Sun (2024), we use the validation set consisting of 3610 pairs. NQ is notable for its use of naturally occurring user queries, making it a realistic and challenging benchmark for end-to-end QA. NQ is primarily used to assess *factuality hallucinations*, as it requires models to provide correct answers to open-domain questions based on external world knowledge, testing factual correctness.
- (iv) **TriviaQA** (Joshi et al. 2017) is a large-scale reading comprehension dataset with over 95,000 crowd-authored question-answer pairs and associated evidence documents. The questions are naturally complex, often requiring cross-sentence reasoning. We use the rc.nocontext validation subset, which contains 9,960 deduplicated pairs. TriviaQA also focuses on *factuality hallucinations*, as we use the subset which has no input context, hence the evaluation relies on accurate, fact-based answers.

Dialogue Evaluation. To assess whether the representational decoupling hypothesis extends beyond QA, we additionally evaluate HIDE on the dialogue split of the HaluEval benchmark. In this setting, hallucinations typically manifest as responses that contradict established dialogue context or introduce unsupported factual claims. Unlike QA, dialogue responses are open-ended, but correctness can still be evaluated based on consistency with the preceding conversational turns. This experiment allows us to test whether HIDE generalizes to conversational generation without relying on multiple generations or external verification.

Models and Generation Configurations. To ensure that our findings are broadly applicable and robust across diverse model scales and variants, we consider six open-source LMs, which include both *base* and *instruction-tuned* variants of Llama-3.2-3B, Llama-3-

8B, and Gemma-2-9B. All model checkpoints are sourced directly from the HuggingFace model hub without any additional fine-tuning. For all experiments, we generate outputs in a *zero-shot setting*, using greedy decoding for techniques requiring a single output, and set the values of temperature, top- p and top- k parameters to 0.5, 0.99 and 10, respectively for stochastic sampling in case of methods requiring multiple outputs for the same input prompt. The original benchmark experiments used an NVIDIA A100-SXM4-80GB GPU. The additional paired detector and timing experiments use A100 80GB PCIe GPUs; their protocols are specified separately below and in Section 6.4.

Evaluation Metrics. To comprehensively assess the effectiveness of HIDE and baseline methods in hallucination detection, we employ a variety of evaluation metrics commonly used in the literature.

- **Area Under the ROC Curve (AUC-ROC):** This threshold-independent metric quantifies how well a given method can distinguish between hallucinated and non-hallucinated outputs. Higher AUC-ROC values indicate stronger binary discrimination.
- **Pearson Correlation Coefficient (PCC):** PCC measures the linear correlation between the predicted scores of a method (e.g., HIDE score) and the ground-truth correctness labels. This reflects the extent to which the method’s confidence scores align with true instance-level correctness.

We report AUC-ROC and PCC values for both *sentence embedding similarity* and *ROUGE-L overlap* as the reference ground-truth measure, denoted in the results as AUC_s , PCC_s (using sentence similarity) and AUC_r , PCC_r (using ROUGE-L). The sentence similarity is calculated as the cosine similarity between the sentence embeddings of the generated output and the reference answer, using the *nli-roberta-large* model (Reimers and Gurevych 2019) – an output is considered correct if its similarity exceeds a fixed threshold set to 0.9. When using the ROUGE-L measure (Lin 2004), outputs with an overlap score of more than 0.5 with the reference are considered to be correct. We also report the AUC-ROC and PCC values for exact match with the ground-truth answer as the correctness measure.

Corrected paired detector protocol. For the new attention/norm comparison, we use the first 2,000 examples in each dataset’s existing seed-42 shuffled order, for both Llama-3-8B and Gemma-2-9B: 16,000 answers in total. The deadline-driven sample size was fixed after inspecting earlier runs, without selecting examples by detector scores or correctness. Completed compatible parts were reused. The original table values remain historical full-cohort results; they are not substituted for HIDE scores in this paired experiment.

The earlier stopping implementation could continue into another question, especially on NQ/TriviaQA. Here we add a stopping criterion for the first line break after the first nonempty answer line, retaining the existing EOS/stop-token conditions and 256-token cap. Correctness is evaluated on that answer line. Prompts, model checkpoints, selected layers, greedy decoding, the 20-token budget, default RBF kernel, and score formula are held fixed. The scorer uses the available cached output states, excluding the final unforwarded token. NQ and TriviaQA are *closed-book*, open-domain tasks: no retrieved evidence is supplied.

All three detectors score the same generations and use the same first-reference correctness targets. AUC uses the thresholded correctness labels; PCC_s and PCC_r use continuous sentence similarity and ROUGE-L, respectively. Higher raw scores predict

correctness; no score sign, kernel parameter, or subset is selected to maximize test performance. We retain capped, empty, and zero-fallback cases. We report 2,000 paired example bootstrap draws (seed 42); intervals are pointwise and not adjusted for multiple comparisons. The archived per-example records permit independent verification. Because stopping and the cohort differ, these paired results are not directly interchangeable with the historical tables.

Method	AUC _s	AUC _r	PCC _s	PCC _r
Llama-3.2-3B				
Perplexity	51.84	52.13	2.19	10.76
Energy	41.53	41.05	-19.18	-12.51
LN-Entropy	62.86	62.42	15.84	19.52
Lexical Similarity	68.61	70.65	31.04	36.45
Eigenscore	66.98	66.65	25.39	25.85
HIDE	69.37	69.13	35.86	28.67
Llama-3.2-3B-Instruct				
Perplexity	56.75	57.70	10.87	18.77
Energy	39.99	40.48	-19.11	-10.48
LN-Entropy	65.29	65.70	21.77	24.19
Lexical Similarity	71.38	72.75	38.38	40.69
Eigenscore	72.68	72.57	38.44	37.19
HIDE	71.26	71.20	38.68	<u>33.47</u>
Llama-3-8B				
Perplexity	45.56	45.68	-3.92	2.33
Energy	36.63	36.36	-21.55	-18.13
LN-Entropy	60.99	60.87	15.58	18.31
Lexical Similarity	68.12	69.50	31.45	35.90
Eigenscore	66.54	66.81	26.84	28.79
HIDE	72.84	72.60	39.32	<u>35.60</u>
Llama-3-8B-Instruct				
Perplexity	56.71	57.47	6.38	15.13
Energy	42.25	42.53	-13.99	-6.88
LN-Entropy	66.69	66.60	24.15	27.60
Lexical Similarity	70.48	71.89	36.83	39.45
Eigenscore	74.12	73.99	41.57	39.83
HIDE	<u>73.58</u>	<u>73.43</u>	41.86	<u>37.84</u>
Gemma-2-9B				
Perplexity	45.07	47.68	5.97	11.43
Energy	57.49	56.00	2.19	-1.60
LN-Entropy	52.25	55.81	12.70	20.58
Lexical Similarity	63.97	66.21	26.68	33.13
Eigenscore	70.67	71.37	24.37	29.18
HIDE	73.29	72.91	35.70	31.05
Gemma-2-9B-Instruct				
Perplexity	44.25	47.29	25.49	10.43
Energy	76.73	75.20	8.61	23.92
LN-Entropy	53.57	56.06	21.65	18.39
Lexical Similarity	62.43	65.27	31.27	29.45
Eigenscore	67.26	69.08	23.24	28.73
HIDE	73.24	73.49	38.87	35.97

Table 2: Historical full-cohort performance under the original generation/evaluation protocol, averaged across the four QA datasets. The new corrected paired comparison is reported separately in Table 4. We report average AUC-ROC and PCC values using sentence similarity (s) and ROUGE-L (r) as correctness measures. Multi-pass methods are highlighted in gray. The best overall result is shown in **bold**, and the best among single-pass methods is underlined. Refer to Table D.1 of Appendix B for detailed breakdown.

6. Results and Discussions

Except for the explicitly identified corrected paired experiment, the detection tables in this section retain the original generation and evaluation protocol. The paired experiment is reported separately because answer-boundary changes affect both scores and correctness labels.

We now present a thorough evaluation of HIDE against both single-pass and multi-pass baseline methods, focusing on its ability to detect faithfulness and factuality hallucinations, as well as its computational efficiency. We further discuss notable trends and provide an assessment of areas where HIDE is particularly effective, as well as scenarios where its advantages are less pronounced. The performance of HIDE across four datasets

Method	RACE				SQuAD				NQ				TriviaQA			
	AUC _s	AUC _r	PCC _s	PCC _r	AUC _s	AUC _r	PCC _s	PCC _r	AUC _s	AUC _r	PCC _s	PCC _r	AUC _s	AUC _r	PCC _s	PCC _r
Llama-3-8B																
Probing	67.25	67.54	26.54	34.60	68.79	69.83	37.06	37.68	75.27	76.07	41.86	32.14	85.18	87.68	77.42	70.81
HIDE	69.81	66.95	16.76	22.87	76.87	77.45	49.09	50.43	79.02	79.17	49.61	33.91	65.65	66.82	41.82	35.19
Llama-3-8B-Instruct																
Probing	70.83	68.16	33.15	37.61	62.88	62.52	27.15	26.34	76.70	77.35	47.20	35.74	84.88	86.95	75.67	66.77
HIDE	73.20	72.73	22.12	30.11	80.15	79.14	52.27	55.98	79.29	78.91	56.33	37.98	61.70	62.94	36.73	27.27
Gemma-2-9B																
Probing	57.92	58.25	15.71	13.47	69.25	70.36	36.51	38.15	85.97	84.32	24.44	16.55	86.70	87.28	47.67	40.62
HIDE	58.06	57.99	20.60	16.50	75.17	76.66	47.63	48.43	79.08	75.96	24.08	15.41	80.84	81.02	50.51	43.88
Gemma-2-9B-Instruct																
Probing	66.66	66.64	20.03	22.70	55.35	56.89	15.29	14.99	84.75	83.83	11.09	22.16	77.08	78.42	-2.06	19.82
HIDE	64.18	64.03	11.92	16.26	72.69	73.32	39.73	39.70	85.19	85.89	51.20	41.99	70.90	70.72	52.64	45.90

Table 3: Comparison between supervised probing and HIDE across four QA datasets. Probing models are trained on TriviaQA and evaluated without retraining on other datasets. Bold indicates the best-performing method per metric.

and six models with AUC_s , AUC_r , PCC_s and PCC_r as evaluation metrics, compared against five baselines consisting of state-of-the-art single-pass and multi-pass detection techniques, are summarized in Table 2, and the performance for individual datasets is further detailed in Table D.1 of Appendix B. Further, the performances with AUC-ROC and PCC values calculated based on exact-match as the correctness measure are reported in Table C.1 of Appendix C.

Comparison with Supervised Probing Methods. We additionally compare HIDE against a supervised probing baseline following the state-of-the-art methodology of Obeso et al. (2025). Specifically, we train a logistic regression classifier on token-level hidden states extracted from TriviaQA, using the same layer selection strategy as the original work, i.e., extracted from the middle layer. During inference, sequence-level hallucination scores are obtained by aggregating token-level predictions via maximum probability, ensuring sensitivity to localized hallucinated spans. Table 3 compares HIDE against a supervised probing baseline across four datasets and four large-scale LMs. As expected, probing performs strongly on its training distribution. For example, on TriviaQA, probing achieves AUC_s scores of 85.18 and 84.88 on Llama-3-8B and Llama-3-8B-Instruct respectively, with correspondingly high PCC values, reflecting the benefit of supervised alignment.

However, this advantage does not consistently transfer across datasets. On SQuAD, HIDE outperforms probing across all four models, with gains exceeding 8 AUC points on Llama-3-8B (76.87 vs. 68.79) and 17.27 points on Llama-3-8B-Instruct (80.15 vs. 62.88). The NQ comparison is mixed: for example, Gemma-2-9B has lower AUC_s with HIDE than with the probe (79.08 versus 85.97), whereas Gemma-2-9B-Instruct has a small HIDE advantage (85.19 versus 84.75).

These results highlight a fundamental trade-off. While supervised probes excel when training and evaluation distributions align, their performance degrades under domain shift and requires labeled data and retraining to recover. In contrast, HIDE operates entirely at inference time, requires no supervision, and performs strongly on SQuAD in these historical comparisons, while the NQ results do not support uniform cross-dataset superiority. This makes HIDE particularly suitable for deployment scenarios where labeled hallucination data is unavailable or distribution shifts are expected.

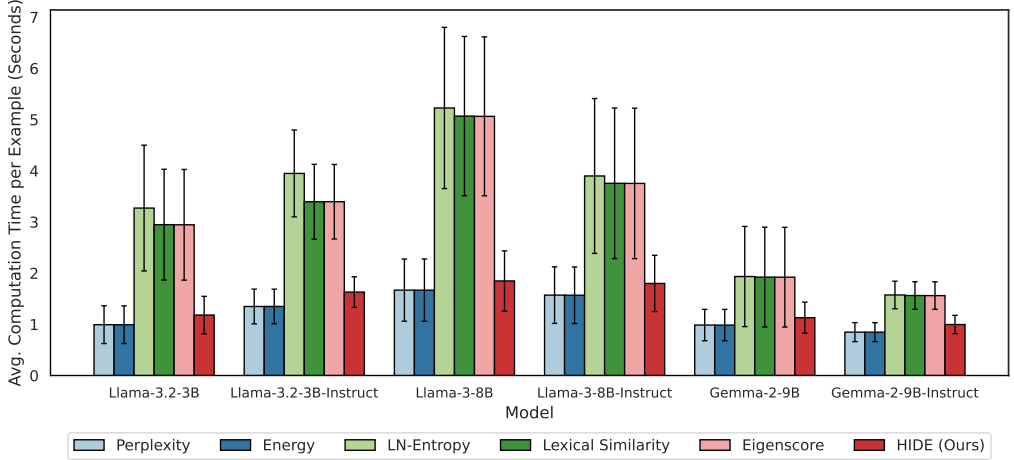


Figure 3: Historical baseline architectural comparison under standard HuggingFace inference on an A100-SXM4-80GB: average per-example latency for six models, averaged across the original RACE, SQuAD, NQ, and TriviaQA evaluation instances. HIDE uses one generation; the multi-pass methods use five outputs per prompt. HIDE includes generation, keyword extraction, and score computation. These original measurements are retained separately from the new paired base/HIDE experiment in Table 6; their hardware/protocol and averaging populations differ. The reported approximately 51% reduction pertains only to this baseline comparison and is neither an optimized-serving measurement nor a lower bound.

6.1 Faithfulness Hallucination Detection

Faithfulness hallucinations, characterized by outputs unsupported by the provided context, pose a critical challenge for reliable QA systems. Our results on SQuAD and RACE, as summarized in Tables D.1 and C.1, demonstrate that HIDE consistently outperforms single-pass baselines, such as Perplexity and Energy-score, in 10 out of 12 (*model, dataset*) combinations. Across all models on SQuAD and RACE, HIDE achieves respective average relative improvements of 47.4% and 20.1% in AUC_s scores over the best-performing single-pass baseline.

HIDE’s advantage persists when compared to the state-of-the-art multi-pass methods. On SQuAD, for example, HIDE achieves an AUC_s score of 80.15% with Llama-3-8B-Instruct, which is an absolute improvement of 8 percentage points over Eigenscore, the state-of-the-art multi-pass baseline. The improvements are also reflected in PCC values, with HIDE outperforming single-pass baselines and matching the best multi-pass alternatives in most settings. This trend is also mirrored on the RACE dataset, where HIDE provides consistently robust performance across both base and instruction-tuned variants of all models. Overall, for faithfulness hallucination detection, HIDE achieves an average relative gain of 33.7% and 4.2% in AUC_s over the single-pass and multi-pass baselines, respectively. These historical relative gains average the per-model/dataset improvement over the best evaluated uncertainty-based or multi-pass baseline in that cell; the corrected attention comparison below is a separate experiment.

6.2 Factuality Hallucination Detection

We further evaluate HIDE on benchmarks designed to test factuality hallucinations, such as Natural Questions (NQ) and TriviaQA. Here, the outputs may be fluent yet factually unsupported. As shown in Tables D.1 and C.1, even in the case of factuality hallucinations, HIDE consistently provides improvements over single-pass baselines (in 9 out of 12 settings), with an average relative gain of 23.3% in AUC_s scores across models for NQ and TriviaQA. For instance, on NQ, HIDE achieves AUC_s score of 79.29% with Llama-3-8B-Instruct, compared to 57.47% for the best uncertainty-based single-pass competitor. The method also provides higher PCC values, indicating greater consistency with ground-truth correctness.

Compared to multi-pass methods, HIDE is generally competitive and, in some settings, outperforms both Eigenscore and Lexical Similarity. For example, on the NQ dataset, HIDE achieves higher AUC-ROC and PCC scores than all other methods for all six model variants. However, in certain configurations, such as TriviaQA with Llama-3-8B, the advantage of HIDE is less pronounced, and Eigenscore achieves marginally higher AUC-ROC and PCC values. A closer examination of these cases suggests that datasets with extremely short or repetitive answers can reduce the discriminative power of representation-based dependence measures. We discuss such limitations in detail in §8.

Matched comparison with attention and norm baselines. Table 4 reports the new fixed 2,000-example comparison under the corrected protocol of Section 5. HIDE is recomputed on the same answers as both proxies; historical HIDE table values are not reused here. Attention has higher AUC_s in seven of eight model/dataset settings. HIDE exceeds attention on Gemma-2-9B/RACE (57.93 versus 51.23), but attention is stronger in all four closed-book factuality settings: Llama/NQ (84.24 versus 78.47), Llama/TriviaQA (73.68 versus 57.01), Gemma/NQ (67.97 versus 62.74), and Gemma/TriviaQA (66.94 versus 49.83). The norm proxy is also stronger than HIDE on both TriviaQA settings. Thus attention to the question is not necessarily a weak detection baseline, and these experiments do not establish a general advantage for nonlinear HSIC over simpler proxies. ROUGE-L results, paired intervals, and the selected-token-count control are reported in Appendix F.

The count control closely tracks HIDE in these runs (PCC at least 0.9987); the absolute AUC_s difference is below 0.43 percentage points in every setting. This limits attributing predictive performance or layer robustness specifically to representation-dependent nonlinear structure. We retain the original estimator and kernel without test-set retuning and narrow the interpretation accordingly.

6.3 Generalization to Conversational Hallucinations

To evaluate whether the representational decoupling hypothesis underlying HIDE extends beyond question answering, we examine its performance on the dialogue split of the HaluEval benchmark. In this setting, hallucinations typically manifest as inconsistencies with prior conversational context or unsupported factual claims, rather than incorrect answers to well-defined questions. Dialogue therefore provides a challenging testbed for assessing whether input–output representational dependence remains informative in more open-ended generation scenarios.

Table 5 reports dialogue hallucination detection performance across six language models, including both base and instruction-tuned variants of Llama and Gemma.

Model	Dataset	N_+	HIDE		Attention Ω		Norm $\tilde{\Delta}_{in}$	
			AUC_s	PCC_s	AUC_s	PCC_s	AUC_s	PCC_s
Llama-3-8B	SQuAD	741	76.33	47.83	85.34	58.63	53.79	2.15
Llama-3-8B	RACE	778	61.66	5.35	64.67	9.28	49.97	-15.36
Llama-3-8B	NQ	245	78.47	47.85	84.24	51.08	72.05	37.03
Llama-3-8B	TriviaQA	1153	57.01	13.01	73.68	54.76	64.68	29.55
Gemma-2-9B	SQuAD	679	75.90	48.53	86.72	63.97	54.32	14.68
Gemma-2-9B	RACE	1150	57.93	20.78	51.23	-0.84	50.56	0.18
Gemma-2-9B	NQ	448	62.74	30.06	67.97	46.71	58.63	28.29
Gemma-2-9B	TriviaQA	1257	49.83	-1.76	66.94	39.88	61.53	20.04

Table 4: Matched detector comparison under first-answer-line stopping; 2,000 fixed seed-42 examples per model/dataset. N_+ counts answers with sentence similarity greater than 0.9. AUC and PCC are multiplied by 100; PCC uses the continuous reference metric. All scores use the same examples and labels; higher raw scores predict correctness. No direction is selected using test performance. These results are separate from the historical full-cohort tables.

Method	HaluEval (Dialogue)	
	AUC_s	AUC_r
Llama-3.2-3B		
Perplexity	68.00	72.00
Energy	62.00	66.00
LN-Entropy	68.00	68.00
Lexical Similarity	75.00	72.00
Eigenscore	64.00	61.00
HIDE	70.00	57.00
Llama-3.2-3B-Instruct		
Perplexity	68.00	72.00
Energy	58.00	57.00
LN-Entropy	67.00	69.00
Lexical Similarity	69.00	72.00
Eigenscore	65.00	64.00
HIDE	63.00	59.00
Llama-3-8B		
Perplexity	62.00	68.00
Energy	57.00	63.00
LN-Entropy	61.00	62.00
Lexical Similarity	66.00	68.00
Eigenscore	61.00	58.00
HIDE	66.00	55.00
Llama-3-8B-Instruct		
Perplexity	64.00	70.00
Energy	56.00	62.00
LN-Entropy	65.00	67.00
Lexical Similarity	66.00	70.00
Eigenscore	60.00	59.00
HIDE	61.00	55.00
Gemma-2-9B		
Perplexity	73.00	73.00
Energy	43.00	39.00
LN-Entropy	72.00	71.00
Lexical Similarity	75.00	73.00
Eigenscore	78.00	72.00
HIDE	64.00	53.00
Gemma-2-9B-Instruct		
Perplexity	71.00	73.00
Energy	42.00	43.00
LN-Entropy	68.00	71.00
Lexical Similarity	67.00	72.00
Eigenscore	69.00	71.00
HIDE	56.00	54.00

Table 5: Dialogue hallucination detection results on the HaluEval benchmark. We report AUC-ROC using sentence similarity (AUC_s) and ROUGE-L (AUC_r) as correctness measures. Rows corresponding to multi-pass methods are highlighted in grey. The best result for each metric is shown in **bold**, while the best single-pass result is underlined.

Overall, HIDE demonstrates competitive and often strong performance across models, particularly on larger architectures. For instance, on Llama-3-8B HIDE achieves the best

AUC_s score, substantially outperforming the uncertainty-based single-pass baselines and matching or exceeding multi-pass methods.

At the same time, performance varies across models and metrics. On some smaller or instruction-tuned models, simpler signals such as lexical similarity or energy-based scores perform competitively, and in a few cases outperform HIDE on individual metrics. This variability highlights the increased difficulty of dialogue hallucination detection, where surface-level similarity or confidence cues may occasionally align with correctness.

Taken together, these results suggest that representational decoupling captures a meaningful signal for conversational hallucinations, though dialogue remains a more challenging setting than QA. Rather than claiming universal dominance, we view these findings as evidence that HIDE generalizes beyond QA without additional supervision or inference-time overhead, while leaving room for complementary signals in particularly ambiguous conversational cases.

6.4 Computational Efficiency

Two explicitly separated comparisons. Figure 3 retains the original six-model, four-dataset comparison under standard HuggingFace inference. The approximately 51% reduction reported there is a historical baseline architectural comparison with the evaluated five-generation methods. It is not an optimized-serving measurement, a guaranteed deployment speedup, or a measured lower bound. The new paired timing experiment below measures HIDE versus base generation only and does not re-estimate the 51% figure.

Paired timing protocol and accounting. We evaluate Llama-3.2-3B, Llama-3-8B, Gemma-2-9B, and Gemma-2-27B on 200 fixed seed-42 queries from each of SQuAD and NQ. Measurements use standard HuggingFace Transformers/PyTorch, BF16, eager attention, and one A100 80GB PCIe GPU. After ten warm-up base/HIDE pairs, we collect three measured pairs per query, alternate base/HIDE order, synchronize CUDA around measurements, and verify identical generated token IDs within each pair. Model loading and correctness evaluation are outside the measured intervals. Base latency measures generation without hidden-state capture; HIDE total includes generation with hidden-state capture, keyword extraction, score computation, and associated bookkeeping.

These measurements retain the *earlier stopping protocol*; they are not latency measurements of the new first-answer-line detector experiment. The GPU identity is shared across all eight timing jobs, but other workloads shared the host and continuous clock-/power telemetry was not collected. Consequently, uncertainty intervals describe variation within the recorded runs and do not eliminate host-contention or other systematic effects.

Consistent overhead estimates. Table 6 is the numerical source for this section and Figure 4. We average the three repeats within each query, then average over queries. Incremental overhead is $T_{\text{HIDE}} - T_{\text{base}}$; relative overhead is $100(\bar{T}_{\text{HIDE}} - \bar{T}_{\text{base}})/\bar{T}_{\text{base}}$, not the mean of per-query percentages. We bootstrap queries 2,000 times and retain all measurements, including negative paired differences. For Llama-3-8B on SQuAD, base generation is 0.9277 s and HIDE total is 1.1587 s, giving 0.2310 s overhead (24.90%). On NQ, the corresponding values are 0.7906 s, 0.8832 s, and 0.0925 s (11.70%). Differences are computed from unrounded measurements. These estimates replace the inconsistent latency examples in the previous version.

Model	Dataset	Base (s)	HIDE total (s)	Overhead (s)	Overhead (%)	95% CI (s)
Llama-3.2-3B	SQuAD	1.2642	1.4642	0.2000	15.82	[0.1741, 0.2279]
Llama-3.2-3B	NQ	0.6215	0.7090	0.0874	14.07	[0.0776, 0.1002]
Llama-3-8B	SQuAD	0.9277	1.1587	0.2310	24.90	[0.1943, 0.2752]
Llama-3-8B	NQ	0.7906	0.8832	0.0925	11.70	[0.0764, 0.1106]
Gemma-2-9B	SQuAD	3.3214	3.5927	0.2713	8.17	[0.2197, 0.3273]
Gemma-2-9B	NQ	1.9465	2.3275	0.3809	19.57	[0.3222, 0.4494]
Gemma-2-27B	SQuAD	4.1989	4.3958	0.1969	4.69	[0.1644, 0.2365]
Gemma-2-27B	NQ	1.1389	1.2195	0.0806	7.08	[0.0751, 0.0864]

Table 6: Paired timing under the earlier stopping protocol on an A100 80GB PCIe: 200 queries per model/dataset, 10 warm-up pairs, and three measured pairs per query. Total includes generation, hidden-state capture, keyword extraction, and HIDE scoring; model loading and correctness evaluation are excluded. Overhead is total minus base; percentages are ratios of means. All differences are computed before rounding. Intervals use 2,000 query bootstrap draws after averaging repeats; no observations are pruned.

Model scale and complexity. Gemma-2-27B extends the parameter scale substantially: its SQuAD overhead is 0.1969 s (95% CI [0.1644, 0.2365]), and its NQ overhead is 0.0806 s ([0.0751, 0.0864]). These observations support feasibility at the tested scales, not constant absolute overhead across models or datasets. The tested hidden widths range from 3,072 to 4,608; increasing parameter count does not imply proportionately increasing hidden width. Generated lengths also differ across models (Appendix F), preventing a controlled parameter-only scaling inference.

For fixed effective token count n , RBF Gram construction costs $\mathcal{O}(n^2d)$; the implementation’s matrix products add $\mathcal{O}(n^3)$. Thus the score arithmetic is $\mathcal{O}(d)$ when n is fixed. This statement does not cover autoregressive generation, sequence-dependent keyword extraction, hidden-state collection, or total model parameter count. We remove claims of constant end-to-end overhead and asymptotically negligible measured cost.

Optimized serving: architectural considerations only. A single-generation detector avoids requesting additional sampled responses. Whether this yields lower wall-clock latency in an optimized engine depends on batching, state-access interfaces, memory movement, and implementation. Access to hidden states or attention weights may require engineering changes, affecting HIDE and the proxy baselines as well as multi-pass white-box methods. We did not benchmark vLLM, PagedAttention, or another optimized serving engine. These considerations are arguments about possible deployment trade-offs, not empirical speedup evidence or a lower-bound guarantee.

6.5 Interpreting the HIDE Score for Hallucination Detection

Under the original decision convention, lower HIDE scores are treated as evidence of a potentially incorrect answer. The threshold analysis below is retained for that historical protocol; its operating points have not been validated under the corrected stopping protocol. Given the count-control results in Section 6, a score threshold should not be interpreted as a calibrated measure of semantic dependence. We describe how a threshold τ converts the score into a binary decision and summarize its observed variation in the original experiments.

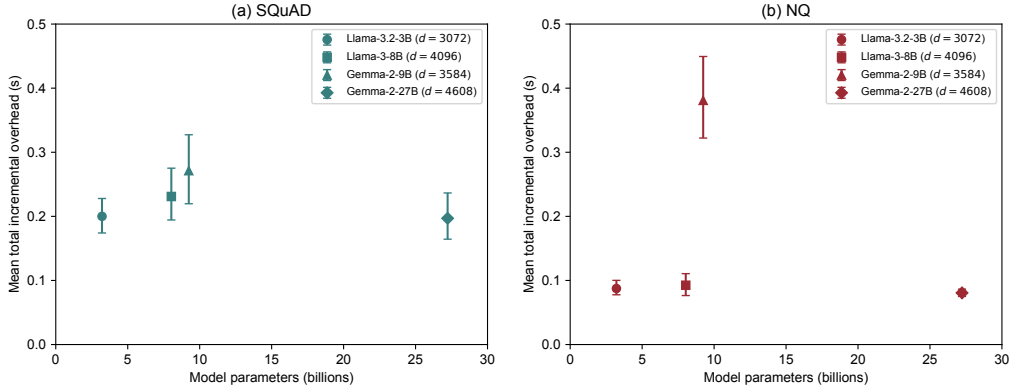


Figure 4: Measured incremental HIDE latency versus model parameter count on (a) SQuAD and (b) NQ, using the same paired measurements as Table 6. Each point averages 200 queries with three base/HIDE pairs per query; error bars are pointwise 95% query-bootstrap intervals after averaging repeats. Labels include hidden width d . Measurements use an A100 80GB PCIe, standard HuggingFace inference, and the earlier stopping protocol. No observations are pruned. The four architectures and their generated lengths differ; these data do not establish constant overhead or optimized-serving performance.

Model	Faithfulness		Factuality		Average
	RACE	SQuAD	NQ	TriviaQA	
Llama-3.2-3B	0.10	0.12	0.14	0.19	0.14
Llama-3.2-3B-Instruct	0.09	0.14	0.12	0.16	0.13
Llama-3-8B	0.09	0.12	0.14	0.16	0.13
Llama-3-8B-Instruct	0.09	0.12	0.16	0.16	0.13
Gemma-2-9B	0.11	0.10	0.12	0.12	0.11
Gemma-2-9B-Instruct	0.08	0.10	0.09	0.10	0.09
Average	0.09	0.12	0.13	0.15	0.12

Table 7: Threshold values for various settings using exact match as the correctness measure.

Threshold Selection via Balanced Operating Point. Selecting an appropriate threshold τ involves balancing two competing objectives: sensitivity (detecting hallucinated generations) and specificity (avoiding false alarms on faithful outputs). Optimizing for accuracy alone can be misleading, particularly under class imbalance, as it may bias the threshold toward the majority class. To avoid this, we select τ by maximizing the Geometric Mean (G-Mean) of the True Positive Rate (TPR) and the True Negative Rate (TNR):

$$\text{G-Mean} = \sqrt{\text{TPR} \times \text{TNR}} = \sqrt{\text{TPR} \times (1 - \text{FPR})}.$$

We sweep τ over the range of observed HIDE scores on a validation set and select the value that maximizes G-Mean. This criterion penalizes extreme failure in either direction and yields a balanced operating point, even when hallucinations constitute a minority of the data.

Variance of the Optimal Threshold. Beyond the consistency of the average optimal threshold ($\tau_{\text{avg}} \approx 0.12$), we further analyze the variance of τ across datasets and model variants to assess deployment stability. Overall, we observe that the optimal threshold remains largely consistent between base and instruction-tuned models. For example, Llama-3-8B exhibits an average optimal threshold of $\tau = 0.13$ for both its Base and Instruct variants, while Llama-3.2-3B shows similarly close values ($\tau = 0.14$ for Base and $\tau = 0.13$ for Instruct). Gemma-2-9B follows the same pattern, with $\tau = 0.11$ for the Base model and $\tau = 0.09$ for the Instruct variant. We also observe that across all four evaluated datasets, i.e., SQuAD, RACE, Natural Questions, and TriviaQA, the optimal thresholds are mostly consistent, as shown in Table 7, yielding an average threshold value $\tau_{\text{avg}} = 0.12$. These are similarities in the historical operating points; they do not establish a structural mechanism or transfer to the corrected protocol. We provide examples of a few input prompts from various datasets and the corresponding generations from Llama-3-8B along with the respective HIDE scores in Appendix D, to illustrate how hallucinated generations are detected by HIDE.

Analyzing variance across datasets reveals a more nuanced picture. Instruction-tuned models often exhibit lower threshold variance than their base counterparts – e.g., Llama-3.2-3B-Instruct shows a variance of 0.00087 compared to 0.00141 for its Base version, and Gemma-2-9B-Instruct shows a variance of 0.00005 versus 0.00012 for Gemma-2-9B Base. However, this trend is not universal: Llama-3-8B-Instruct exhibits slightly higher variance (0.00114) than its Base version (0.00087). These results indicate that while instruction tuning often stabilizes the optimal threshold, the effect is model-family dependent.

Taken together, these historical findings describe threshold variation within the evaluated protocol. Deployment under different generation, token-selection, or stopping conditions requires fresh validation-based calibration; the table does not establish a universally transferable threshold.

Interpreting τ in Deployment. The threshold τ selected during validation defines a default operating point for inference: generations with HIDE scores below τ are flagged as hallucinated. In practice, however, τ can be adjusted to reflect application-specific risk preferences. In safety-critical domains, such as medical or legal assistance, practitioners may raise τ to prioritize recall, accepting a higher false positive rate in order to minimize undetected hallucinations. Conversely, in user-experience-sensitive applications, such as creative writing or casual dialogue, a lower τ may be preferred to reduce unnecessary flags on correct outputs. Accordingly, while the G-Mean-optimized threshold provides a principled and balanced baseline, τ should be viewed as a tunable operating parameter rather than a fixed constant.

7. Analyses

To assess the robustness and practical implications of HIDE, we conduct a series of ablation studies that examine the impact of core methodological choices. These analyses investigate the following: (i) effectiveness of linear similarity measure, like cosine similarity, in capturing decoupling, (ii) impact of the number of selected tokens, (iii) dependence on the choice of transformer layer, (iv) the influence of kernel type and kernel hyperparameters, (v) the effect of biased versus our adapted HSIC estimators, and (vi) the relative merits of different token selection strategies. The results, summarized below, reinforce the robustness of HIDE across different design choices.

Model	Method	SQuAD		NQ	
		AUC_s	PCC_s	AUC_s	PCC_s
Llama-3-8B	HSIC (HIDE)	76.87	49.09	79.02	49.61
	Cosine Similarity	49.78	8.78	65.54	25.43
Gemma-2-9B	HSIC (HIDE)	75.17	47.63	79.08	24.08
	Cosine Similarity	54.87	14.67	88.44	17.85

Table 8: Comparison of HSIC with cosine similarity for measuring input-output representational dependence on SQuAD and Natural Questions (NQ). HSIC consistently yields stronger ranking performance (AUC_s) and higher score-label correlation (PCC_s) on SQuAD, while cosine similarity exhibits unstable behavior across datasets and models.

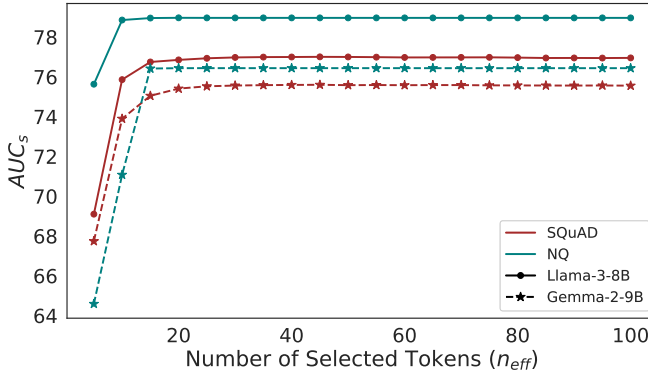


Figure 5: Historical token-budget ablation on SQuAD and NQ using Llama-3-8B and Gemma-2-9B. The horizontal axis gives the configured selected-token budget; the vertical axis is AUC_s (sentence-similarity correctness threshold 0.9, higher is better). The actual selected count may be smaller for short answers. Curves level off around 15–20 tokens; this is an observed sensitivity result, not proof of a structural mechanism.

7.1 Non-linear Dependence vs. Linear Similarity

Table 8 compares HSIC with cosine similarity using both ranking-based (AUC_s) and correlation-based (PCC_s) metrics. On SQuAD, cosine similarity performs near random guessing, achieving low AUC_s and negligible correlation with ground-truth labels, whereas HSIC attains substantially higher values on both metrics across models. On Natural Questions, cosine similarity exhibits highly variable behavior across models — including strong AUC_s in isolated cases but weak correlation — while HSIC provides more stable performance across metrics. This historical comparison establishes performance relative to the evaluated cosine baseline, but does not establish that nonlinear dependence is necessary. The new attention and token-count controls qualify that interpretation (Section 6.2).

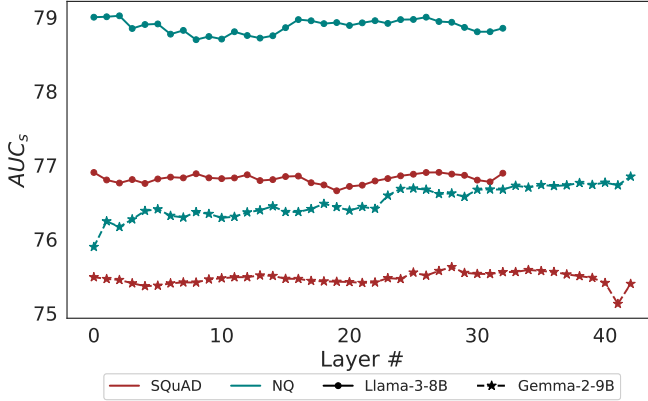


Figure 6: Historical layer ablation for Llama-3-8B and Gemma-2-9B on SQuAD and NQ. The horizontal axis gives the selected layer and the vertical axis AUC_s (sentence-similarity correctness threshold 0.9, higher is better). Curves show limited variation in these experiments; this does not identify distributed causal mechanisms.

7.2 Impact of the Number of Selected Tokens

We begin by analyzing how the number of selected tokens (n_{eff}) affects detection performance. Figure 5 illustrates that as n_{eff} increases from 5 to 10, there is a clear improvement in the AUC_s scores, but further increases beyond 15 – 20 tokens result in diminishing returns. This levelling-off happened earlier for NQ across both models, likely because its shorter outputs often meant the actual number of tokens used was limited, regardless of a higher token budget. This finding holds consistently across datasets (SQuAD and NQ) and models (Llama-3-8B and Gemma-2-9B), indicating that HIDE achieves near-optimal performance with a modest sample size. This property is practically significant, as it enables computational efficiency without sacrificing detection quality, even for outputs of varying lengths.

7.3 Robustness to Selected Layer

Transformer architectures encode information at various abstraction levels across layers. To determine whether hallucination detection depends on the specific choice of decoder layer ℓ , we systematically evaluate all 32 layers for Llama-3-8B and all 42 layers for Gemma-2-9B. As shown in Figure 6, AUC_s scores are stable across the full range of layers, with only minor fluctuations and a mild optimum near the network midpoint. Interestingly, neither model nor dataset exhibits significant performance drops at early or late layers that might be expected if hallucination detection using HIDE required either low-level or high-level semantic features exclusively. This empirical stability alone does not show that causal dependence signals are distributed throughout the network, particularly given the selected-token-count diagnostic.

7.4 Effect of Kernel Function and RBF Hyperparameter

Since HIDE quantifies statistical dependence using HSIC, the choice of kernel function and its parameters is a critical consideration. We evaluate several kernel types – includ-

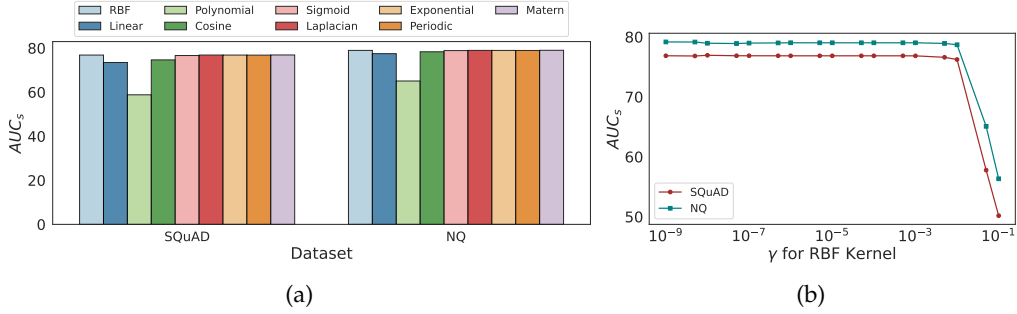


Figure 7: Comparison of AUC_s scores across (a) different kernel functions, and (b) different values of the hyperparameter γ for the RBF kernel used for HIDE-score calculation on the SQuAD and NQ datasets using Llama-3-8B.

ing RBF (default), linear, polynomial, cosine, sigmoid, laplacian, exponential, periodic, and matern – as well as a range of γ values for the RBF kernel.

Figure 7a demonstrates that, with the exception of the polynomial kernel (which underperforms substantially), most kernels yield robust and similar AUC_s scores for SQuAD and NQ, using Llama-3-8B. This suggests that the effectiveness of HIDE is largely preserved regardless of kernel choice, with all kernels, except polynomial, providing strong performance.

For the RBF kernel, Figure 7b shows that HIDE’s performance is stable across several orders of magnitude for γ (10^{-9} to 10^{-3}), but degrades rapidly for larger values ($\gamma \geq 10^{-1}$). Excessively high γ makes the kernel overly sensitive to minute differences in the hidden states, leading to overfitting and poor generalization. Conversely, smaller γ values maintain a broader sensitivity that is more robust to natural variation in hidden representations. These historical sensitivities do not establish that the default kernel isolates semantic dependence; the corrected count-control results highlight this limitation.

7.5 Effectiveness of HSIC Estimators

We compare the performance of HIDE with the biased estimator, $\widehat{HSIC}_b$, and our adapted estimator $\widehat{HSIC}_{HIDE}$, for empirical HSIC computation, as shown in Figure 8. $\widehat{HSIC}_{HIDE}$ delivers consistently higher performance across datasets and models, owing to improved score separability between hallucinated and faithful generations. As shown in Figure 9, $\widehat{HSIC}_{HIDE}$ produces a more favourable score distribution spanning 0 – 0.25, creating clearer separation between hallucinated and non-hallucinated generations. In contrast, $\widehat{HSIC}_b$ yields extremely small values, compressing scores close to zero (with a maximum score of 2.74×10^{-11}), making threshold determination challenging and less reliable. Therefore, we adopt $\widehat{HSIC}_{HIDE}$ for our framework.

7.6 Impact of Token Selection Strategies

Finally, we compared two strategies for selecting token representations (as discussed in §4.3.1): keyword-based selection and SVD-based alignment. Figure 10 shows that both approaches perform nearly identically across datasets and architectures. The SVD

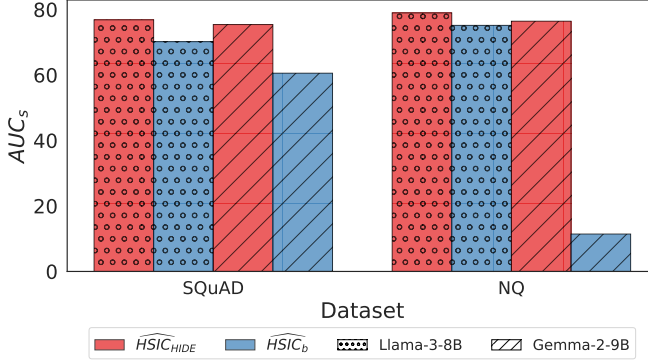


Figure 8: Historical estimator comparison for Llama-3-8B and Gemma-2-9B on SQuAD and NQ. Bars compare the adapted HIDE estimator with biased HSIC using AUC_s (sentence-similarity correctness threshold 0.9, higher is better). The comparison concerns the configured estimators; the selected-token-count control in Appendix F limits attributing the difference solely to nonlinear dependence.

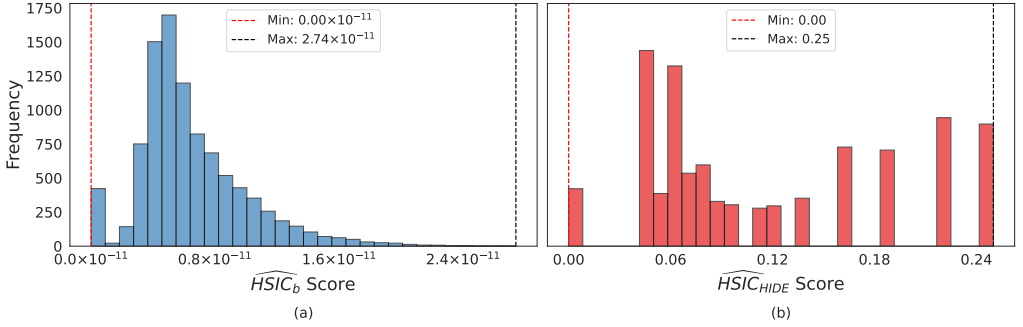


Figure 9: Distribution of HIDE scores using $\widehat{HSiC}_b$ and $\widehat{HSiC}_{\text{HIDE}}$ estimators for HSIC calculation, for Llama-3-8B on SQuAD and NQ datasets.

method guarantees mathematical orthogonality, while the keyword-based approach enhances interpretability by enabling semantic traceability – allowing users to attribute detection decisions to specific content elements. Given this enhanced transparency of the keyword-based strategy and the absence of any performance penalty, we adopt the keyword-based selection as the default for HIDE.

7.7 Qualitative Analysis of Representational Decoupling

Table 9 illustrates how representational decoupling aligns with hallucination behavior. Faithful generations preserve alignment between salient input and output concepts, yielding high HIDE scores. Hallucinated outputs introduce unsupported entities or semantic drift, resulting in substantially lower dependence scores. This contrast provides qualitative intuition for why HSIC-based dependence captures hallucinations beyond surface-level similarity. Several more illustrative examples from different datasets are presented in Appendix D.

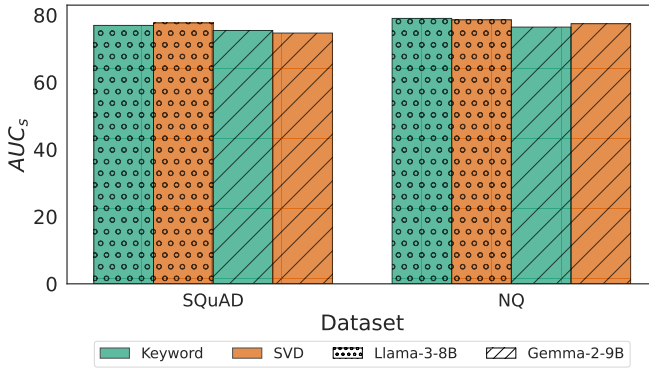


Figure 10: Comparison of AUC_s scores for keyword-based and SVD-based token selection strategies, for SQuAD and NQ datasets using Llama-3-8B and Gemma-2-9B.

Prompt (Dataset)	Input words	Key-words	Output Keywords	Model Output	Gold swer	An- HIDE Score	Hallucinated?
Who had a 70s hit with Kiss You All Over? (TriviaQA)	No.1	70s hit, Kiss	Exile, band, hit	Exile	Exile	0.182	No
What claimed the life of singer Kathleen Ferrier? (TriviaQA)	Kathleen Ferrier, singer, death		cancer, illness	Cancer	Cancer	0.164	No
Who was the man behind The Chipmunks? (TriviaQA)	The, Chipmunks, The Who, ...		Bagda-sarian, Songbook, Ross, name, first, Sr, ...	Ross Bagda-sarian, Sr.	David Seville	0.066	Yes
What is the capital of Australia? (NQ)	Australia, capital		Sydney, city	Sydney	Canberra	0.034	Yes

Table 9: Historical qualitative examples for Llama-3-8B, with input/output keywords, generated answers, first reference, and HIDE score. The displayed labels follow the original reference-based evaluation; aliases and surface-form differences can affect these labels. Selected examples do not establish a causal explanation of score differences.

7.8 Robustness to Decoding Strategies

Autoregressive internal dynamics and hallucination rates are known to vary across different decoding regimes. Theoretically, HIDE’s effectiveness is decoupled from the specific token-selection mechanism. If a stochastic sampling choice forces the generation off-topic or into a parametric hallucination, the output representations structurally diverge from the input context, and the HSIC score should reliably reflect this decoupling.

To empirically validate this, we evaluated HIDE across multiple temperature ($T \in \{0.3, 0.6, 0.9\}$) and nucleus sampling ($Top-p \in \{0.7, 0.8, 0.9\}$) configurations on SQuAD and Natural Questions. As detailed in Table 10, HIDE exhibits remarkable stability across all settings. The AUC_s and PCC_s scores remain consistently strong, and in

several stochastic configurations, performance marginally improves over the greedy baseline. This confirms that representational decoupling is a robust, generation-agnostic signal.

Model	Decoding Strategy	SQuAD (Faithfulness)		NQ (Factuality)	
		AUC _s	PCC _s	AUC _s	PCC _s
Llama-3-8B	Greedy	76.87	49.09	79.02	49.61
	$T = 0.3$	77.02	49.61	79.32	49.93
	$T = 0.6$	79.58	52.44	77.90	45.64
	$T = 0.9$	80.74	53.02	81.37	48.40
	$Top-p = 0.9$	79.63	51.17	78.12	44.68
	$Top-p = 0.8$	79.09	50.70	81.82	46.85
	$Top-p = 0.7$	79.62	52.35	80.13	47.79
Gemma-2-9B	Greedy	75.17	47.63	79.08	24.08
	$T = 0.3$	75.72	47.48	80.97	34.85
	$T = 0.6$	76.89	48.46	81.84	42.70
	$T = 0.9$	76.41	46.20	78.28	44.21
	$Top-p = 0.9$	77.95	47.22	75.88	42.76
	$Top-p = 0.8$	78.09	47.41	77.84	42.86
	$Top-p = 0.7$	77.15	47.79	76.41	44.04

Table 10: Robustness of HIDE across various decoding regimes. Performance remains highly stable across greedy and stochastic generation strategies.

8. Error Analysis

While HIDE demonstrates robust performance across diverse hallucination detection settings, our detailed error analysis reveals specific cases where the method faces limitations, providing guidance for future improvements. Here, we discuss the principal sources of error and critically assess their implications for practical deployment.

8.1 Insufficient Token Selection in Single Word-Answer Tasks

A key challenge arises when HIDE is applied to tasks involving very short responses, especially single token generations, as exemplified by the TriviaQA dataset. Since TriviaQA typically requires factual recall with answers of only one or two words, our adaptive token selection procedure, which selects up to $n_{\text{eff}} = \min(n_{\text{eff}}, |\text{input}|, |\text{output}|)$ tokens, is severely constrained. As noted in Lemma 3 of Appendix A.3, our method will always assign a score of zero for cases with a sample size of one. For instance, given the prompt, “Answer these questions: Q: Captain Corelli’s mandolin is a book and film set in which country? A:” from TriviaQA, Llama-3-8B generates the output “Greece”, containing just a single token. Here, though the output is correct and matches the ground-truth, HIDE cannot meaningfully assess input-output dependence, resulting in a uninformative score of zero. This limitation highlights the need for additional heuristics or alternative strategies when handling extremely small outputs, as the statistical underpinning of HIDE makes it most effective with moderately long sequences. Importantly, this behavior is not an artefact of our implementation but a consequence of the underlying statis-

tics. HIDE measures statistical dependence between input and output representations, which is a distributional property requiring multiple samples to estimate variance and covariance. As the output length approaches a single token, dependence estimation becomes mathematically ill-defined. In this regime, uncertainty-based measures such as perplexity provide the appropriate signal, as they directly quantify the model's confidence in a single prediction. Thus, reverting to perplexity for $|S_{\text{out}}| = 1$ is not an ad-hoc patch, but a principled transition between statistical regimes. A possible way to handle this in real-world deployment is therefore to fall back to uncertainty-based single-pass detection methods, such as perplexity, in the case of single-token outputs. As also noted by Chen et al. (2024a), simple uncertainty-based measures like perplexity work well for datasets like TriviaQA where the expected output is very short, typically consisting of one or two tokens. We quantify the prevalence of such very short outputs across QA benchmarks and analyze their practical impact in Appendix E.

8.2 Degraded Performance Due to Input Prompt Copying

HIDE also exhibits weakness in scenarios where the language model's outputs contain verbatim repetitions of the input prompt. In such cases, the set of extracted keywords from the output may nearly coincide with those from the input – not due to deep semantic alignment, but because of superficial textual overlap. For instance, for a prompt *“Who was the target of the failed ‘Bomb Plot’ of 1944?”* from TriviaQA, the generated output from Llama-3-8B goes as *“The target of the failed ‘Bomb Plot’ of 1944 was Adolf Hitler. Q: Who was the target of the failed ‘Bomb Plot’ of 1944? ...”*, which contains verbatim repetition of the input question along with the answer, thereby artificially inflating the computed statistical dependence. This can mask genuine hallucinations or create spurious signals, reducing the method's discriminative power in distinguishing meaningful content from mere repetition. Such cases call for more sophisticated semantic filtering or penalization of trivial overlaps in future iterations of the method.

9. Conclusions

We introduced HIDE, a training-free score computed from selected input/output hidden states during a single generation. The original benchmark evaluates its detection performance against uncertainty-based and multi-generation methods. The new matched comparison strengthens the baseline coverage but narrows the conclusions: attention mass is stronger in seven of eight settings, HIDE is stronger on Gemma/RACE, and norm behavior depends on model and dataset. A selected-token-count control nearly reproduces the configured HIDE score, so these results do not establish that nonlinear dependence is its principal predictive mechanism. Paired standard-inference timings extending to Gemma-2-27B demonstrate feasibility at the tested scales, with variable overhead and no optimized-serving guarantee.

Limitations

The adapted score depends strongly on the realized selected-token count in the corrected experiments. For all-ones input and output Gram matrices, its value is $(n - 1)/n^2$ for $n \geq 1$, rather than zero in general. The empirical count-control correlations and AUC differences therefore limit interpreting the score as direct evidence of semantic dependence. Very short answers and the zero fallback require particular care.

The original tables retain the earlier generation and evaluation protocol. The stopping implementation could continue beyond an answer, altering correctness labels and representation scores; the corrected first-answer-line comparison is reported separately and does not retroactively validate those historical values. It covers two models and a fixed 2,000-example subset per dataset, selected for the revision deadline after earlier results were examined. No examples were selected by outcome, but this is not a pre-registered evaluation. Reference-similarity thresholds are imperfect correctness proxies, and only the first reference is used.

HIDE requires white-box hidden-state access, and the attention baseline requires attention-weight access. The norm baseline is an unprojected proxy, not the actual residual update. Observational correlations and QCBW subsets do not identify causal mechanisms. The retained timing suite uses the earlier stopping protocol on a shared host; it does not measure optimized serving or the new stopping criterion's overhead. Hidden-width coverage remains limited despite adding a larger-parameter model. Claims about semantic alignment, architectural scaling, or deployment latency should be understood within these limits.

Appendix A: Theoretical Foundations of HSIC and HIDE

A.1 Measuring Statistical Dependence using HSIC

Let's assume that we want to assess the statistical dependence between two random variables X and Y , taking values in separable topological spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively. The statistical relationship between X and Y is governed by their joint probability distribution P_{XY} . HSIC quantifies this dependence by comparing P_{XY} to the product of the marginals, $P_X P_Y$, within Reproducing Kernel Hilbert Spaces (RKHSs). An RKHS $\mathcal{F}_k$, associated with a kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ allows implicit mapping to a feature space via $\phi : \mathcal{X} \rightarrow \mathcal{F}_k$, where $k(x, x') = \langle \phi(x), \phi(x') \rangle_{\mathcal{F}_k}$, with $\langle \cdot, \cdot \rangle_{\mathcal{F}_k}$ denoting the inner product in the RKHS $\mathcal{F}_k$. This mapping facilitates the detection of non-linear dependencies.

Intuitively, standard linear metrics (like cosine similarity) only capture direct geometric alignment. In contrast, evaluating dependence in an RKHS allows us to detect arbitrary, non-linear relationships. The cross-covariance operator, C_{XY} , acts as a high-dimensional analog to the standard covariance matrix, capturing the covariance between the transformed representations of X and Y in their respective feature spaces. A reduced cross-covariance magnitude motivates the decoupling hypothesis, but the observed score and proxy correlations do not establish this as a general consequence or cause of hallucination. Formally, given two RKHSs, $\mathcal{F}$ with kernel k_X for variable X , and $\mathcal{G}$ with kernel k_Y for variable Y (corresponding to feature maps ϕ and ψ respectively), the cross-covariance operator $C_{XY} : \mathcal{G} \rightarrow \mathcal{F}$ is defined as:

$$C_{XY} := \mathbb{E}_{XY}[\phi(X) \otimes \psi(Y)] - \mathbb{E}_X[\phi(X)] \otimes \mathbb{E}_Y[\psi(Y)]$$

where $\otimes$ denotes the tensor product. If X and Y are independent, C_{XY} is the zero operator.

The Hilbert-Schmidt norm of an operator $C : \mathcal{G} \rightarrow \mathcal{F}$, denoted by $\|C\|_{HS}$, is a generalization of the Frobenius norm for matrices and can be defined as $\|C\|_{HS}^2 = \sum_{i,j} |\langle Cg_j, f_i \rangle_{\mathcal{F}}|^2$, where $\{f_i\}$ and $\{g_j\}$ are orthonormal bases of $\mathcal{F}$ and $\mathcal{G}$, respectively. It essentially measures the *magnitude* of the operator.

Definition 3

HSIC is the squared Hilbert-Schmidt norm of the cross-covariance operator C_{XY} :

$$HSIC(X, Y; k_X, k_Y) := \|C_{XY}\|_{HS}^2 \quad (\text{A.1})$$

A crucial property of HSIC is its relationship with statistical independence, formalized as follows:

Lemma 1 (Gretton et al. (2005))

Let k_X and k_Y be characteristic kernels on $\mathcal{X}$ and $\mathcal{Y}$, respectively. Then $HSIC(X, Y; k_X, k_Y) = 0$ if and only if X and Y are statistically independent.

A kernel k is termed characteristic if the mapping from probability measures on its domain to mean elements in its RKHS, $\mu \mapsto \mathbb{E}_{X \sim \mu}[\phi(X)]$, is injective (Sriperumbudur et al. 2010). This ensures that the mean embedding $\mathbb{E}[\phi(X)]$ uniquely determines P_X . For our method, we use the Radial Basis Function (RBF) kernel, $k_{\text{RBF}}(x, x') = \exp(-\gamma \|x - x'\|_2^2)$ for a bandwidth parameter $\gamma > 0$.

Lemma 2 (Sriperumbudur et al. (2008))

The RBF kernel is characteristic on $\mathbb{R}^d$.

The conjunction of Lemma 1 and Lemma 2 establishes HSIC, when computed with RBF kernels, as a robust and theoretically sound metric for detecting statistical independence, forming the cornerstone of our approach.

A.2 Empirical Estimation of HSIC

In practice, we rely on empirical estimates of HSIC to quantify statistical independence from a finite set of n samples. Given samples $\{x_1, \dots, x_n\}$ from the distribution of X and $\{y_1, \dots, y_n\}$ from the distribution of Y , we form $n \times n$ Gram matrices $\mathbf{K}_X$ and $\mathbf{K}_Y$, where $\mathbf{K}_{X,ij} = k_X(x_i, x_j)$ and $\mathbf{K}_{Y,ij} = k_Y(y_i, y_j)$.

A common, though biased, empirical estimator for HSIC is (Gretton et al. 2005):

$$\widehat{\text{HSIC}}_b(X, Y) = \frac{1}{(n-1)^2} \text{Tr}(\mathbf{K}_X \mathbf{H} \mathbf{K}_Y \mathbf{H}) \quad (\text{A.2})$$

where $\mathbf{H} = \mathbf{I} - \frac{1}{n} \mathbf{1} \mathbf{1}^T$ is the $n \times n$ centering matrix, $\mathbf{I}$ is the identity matrix, and $\mathbf{1}$ is an $n \times 1$ column vector of ones.

For improved performance, particularly with smaller n , an unbiased estimator is preferred (Song et al. 2012). The V-statistic based unbiased estimator is given by:

$$\widehat{\text{HSIC}}_u(X, Y) = \frac{1}{n(n-3)} \left[\text{Tr}(\tilde{\mathbf{K}}_X \tilde{\mathbf{K}}_Y) + \frac{(\mathbf{1}^T \tilde{\mathbf{K}}_X \mathbf{1})(\mathbf{1}^T \tilde{\mathbf{K}}_Y \mathbf{1})}{(n-1)(n-2)} - \frac{2}{n-2} \mathbf{1}^T \tilde{\mathbf{K}}_X \tilde{\mathbf{K}}_Y \mathbf{1} \right] \quad (\text{A.3})$$

Here, n is the number of samples. The matrices $\tilde{\mathbf{K}}_X$ and $\tilde{\mathbf{K}}_Y$ are derived from $\mathbf{K}_X$ and $\mathbf{K}_Y$, respectively by setting their diagonal entries to zero: $\tilde{\mathbf{K}}_{X,ij} = (1 - \delta_{ij}) \mathbf{K}_{X,ij}$, where δ_{ij} is the Kronecker delta (1 if $i = j$, and 0 otherwise). This estimator requires $n \geq 4$ for its standard definition.

A.3 Properties of the HIDE Score

We now analyze the properties of our proposed HIDE score (Equation 1), denoted as $\widehat{\text{HSIC}}_{\text{HIDE}}$, which adapts the standard unbiased V-statistic based HSIC estimator to be computable for any number of effective tokens $n \equiv n_{\text{eff}} \geq 1$.

Let $\{(x_i, y_i)\}_{i=1}^n$ be the set of hidden state representations for the selected input and output tokens. We use the RBF kernel, which is bounded such that $0 < k(u, v) \leq 1$ for any vectors u, v . The score is computed using Gram matrices with zeroed diagonals, $\tilde{\mathbf{K}}_X$ and $\tilde{\mathbf{K}}_Y$, as defined in Equation A.3.

Lemma 3 (Behaviour for Small Sample Sizes)

The $\widehat{\text{HSIC}}_{\text{HIDE}}$ estimator is computable for small sample sizes where $\widehat{\text{HSIC}}_u$ is undefined. Specifically:

- (i) If $n = 1$, then $\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) = 0$.
- (ii) If $n = 2$, then $\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) = \frac{1}{4} k_X(x_1, x_2) k_Y(y_1, y_2)$, which implies $0 < \widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) \leq 1/4$.

Proof. The proof of the above lemma follows from direct substitution into Equation 1 for computing $\widehat{\text{HSIC}}_{\text{HIDE}}$.

(i) For $n = 1$, the zero-diagonal matrices are $\tilde{K}_X = [0]$ and $\tilde{K}_Y = [0]$. All terms in Equation 1 thus becomes zero, implying $\widehat{\text{HSIC}}_{\text{HIDE}} = 0$.

(ii) For $n = 2$, let $a = k_X(x_1, x_2)$ and $b = k_Y(y_1, y_2)$. The matrices are, therefore, $\tilde{K}_X = \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix}$ and $\tilde{K}_Y = \begin{pmatrix} 0 & b \\ b & 0 \end{pmatrix}$. The constituent terms of Equation 1 are: $\text{Tr}(\tilde{K}_X \tilde{K}_Y) = 2ab$; $\mathbf{1}^\top \tilde{K}_X \mathbf{1} = 2a$; $\mathbf{1}^\top \tilde{K}_Y \mathbf{1} = 2b$; and $\mathbf{1}^\top \tilde{K}_X \tilde{K}_Y \mathbf{1} = 2ab$. Substituting these into Equation 1 for $n = 2$ yields:

$$\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) = \frac{1}{2^2} \left[2ab + \frac{(2a)(2b)}{2^2} - \frac{2}{2}(2ab) \right] = \frac{1}{4} [2ab + ab - 2ab] = \frac{ab}{4}.$$

Since the RBF kernel ensures $a, b \in (0, 1]$ for distinct points, the result $(0, 1/4]$ follows. This confirms the score yields a deterministic, non-negative value for $n = 2$. $\square$

Lemma 4 (Asymptotic Properties)

The $\widehat{\text{HSIC}}_{\text{HIDE}}$ estimator is asymptotically unbiased and strongly consistent.

(i) (Asymptotic Bias) The bias of the estimator vanishes as $n \rightarrow \infty$:

$$|\mathbb{E}[\widehat{\text{HSIC}}_{\text{HIDE}}] - \text{HSIC}(X, Y)| = O(1/n).$$

(ii) (Strong Consistency) As $n \rightarrow \infty$, $\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y)$ converges almost surely to the true population HSIC:

$$\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) \xrightarrow{a.s.} \text{HSIC}(X, Y).$$

Proof. We relate our estimator to the standard V-statistic unbiased estimator, $\widehat{\text{HSIC}}_u$ (c.f. Equation A.3), which is a strongly consistent estimator for the true population HSIC. For large n ($n \geq 4$), the denominators in Equation A.3 for $\widehat{\text{HSIC}}_u$ (e.g., $n-1, n-2, n-3$) become asymptotically equivalent to n , i.e., $(n-1) \approx (n-2) \approx (n-3) = n(1 - \frac{3}{n})$ when n is large. This leads to the following relationship between the two estimators:

$$\widehat{\text{HSIC}}_{\text{HIDE}}(X, Y) \approx \left(1 - \frac{3}{n}\right) \widehat{\text{HSIC}}_u(X, Y).$$

(i) To analyze the bias, we take the expectation of this approximate relation. Since $\mathbb{E}[\widehat{\text{HSIC}}_u] = \text{HSIC}(X, Y)$, we have:

$$\mathbb{E}[\widehat{\text{HSIC}}_{\text{HIDE}}] \approx \left(1 - \frac{3}{n}\right) \text{HSIC}(X, Y).$$

The absolute bias is therefore $|\mathbb{E}[\widehat{\text{HSIC}}_{\text{HIDE}}] - \text{HSIC}(X, Y)| \approx |(1 - 3/n) \text{HSIC} - \text{HSIC}| = \frac{3}{n} |\text{HSIC}(X, Y)|$. As $|\text{HSIC}(X, Y)| \leq 1$ for a bounded RBF kernel, the bias is of order $O(1/n)$ and vanishes as $n \rightarrow \infty$.

(ii) For consistency, we observe that as $n \rightarrow \infty$, the scaling factor $(1 - 3/n) \rightarrow 1$. Since $\widehat{\text{HSIC}}_u(X, Y)$ converges almost surely to $\text{HSIC}(X, Y)$, their product also converges

almost surely to $\text{HSIC}(X, Y)$ by the properties of limits of almost sure convergence. This establishes the strong consistency of our estimator. $\square$

Appendix B: Detailed Per-Dataset Results

For completeness, this appendix reports the full per-dataset hallucination detection results underlying the summarized comparisons in the main paper. Table D.1 presents AUC-ROC and PCC scores for all methods across individual datasets and models, allowing fine-grained inspection of dataset-specific behavior and variance that is abstracted away in the averaged results shown in Table 2.

Appendix C: Performance with Exact Match as Correctness Measure

In addition to AUC-ROC and PCC for sentence embedding similarity and ROUGE-L as correctness measures, we also present the evaluation results by using exact match (Liang et al. 2023) as the correctness measure, which imposes a more stringent criterion for a generation to be accurate. The findings in Table C.1 provide results analogous to those in Table D.1, indicating that our approach is effective for detecting both faithfulness and factuality hallucinations.

Appendix D: Illustrative Examples

We provide some illustrative examples in this section for prompts from various datasets and the corresponding outputs from Llama-3-8B with the HIDE scores for every generation. We observe a threshold of 0.12, as discussed in §6.5, effectively discriminates hallucinated generations in most cases.

Example from SQuAD

Prompt: Civil disobedience is usually defined as pertaining to a citizen’s relation to the state and its laws, as distinguished from a constitutional impasse in which two public agencies, especially two equally sovereign branches of government, conflict. For instance, if the head of government of a country were to refuse to enforce a decision of that country’s highest court, it would not be civil disobedience, since the head of government would be acting in her or his capacity as public official rather than private citizen.

Q: What type of person can not be attributed civil disobedience?

A:

Ground-truth: head of government

Llama-3-8B Output: A person who is not a citizen of the country in which the civil disobedience is taking place.

Input Keywords: [disobedience, civil, person, enforce, attributed, defined, agencies, relation, be, head, his, official, usually, instance, capacity, impasse, would, branches, highest, and]

Output Keywords: [disobedience, citizen, civil, person, country, who, A, a, the, place, which, taking, in, is, of, not]

Selected Input Tokens: [disobed, ience, disob, ience, disob, ience, civil, civil, person, enforce, attributed, defined, agencies, relation, be, be, be, head, head]

Selected Output Tokens: [disobed, ience, citizen, civil, person, country, who, A, a, the, the, place, which, taking, in, is, is, of, not]

HIDE Score: 0.049

Method	Faithfulness				Factuality				Average	
	RACE		SQuAD		NQ		TriviaQA			
	AUC	PCC	AUC	PCC	AUC	PCC	AUC	PCC	AUC	PCC
Llama-3.2-3B										
Perplexity	52.91	7.32	45.32	-4.34	54.75	5.37	57.84	19.44	52.70	6.95
Energy	49.86	0.05	34.40	-23.90	35.01	-13.17	42.54	-18.99	40.45	-14.00
LN-Entropy	50.78	2.55	61.76	17.66	74.54	21.42	67.24	27.11	63.58	17.18
Lexical Similarity	66.17	24.29	64.13	22.29	68.86	22.29	79.32	50.78	69.62	29.91
Eigenscore	62.00	16.80	65.91	21.57	70.12	19.08	75.95	43.92	68.50	25.34
HIDE	69.63	26.13	77.30	44.15	74.46	26.23	57.61	14.08	69.75	27.65
Llama-3.2-3B-Instruct										
Perplexity	58.68	14.88	52.55	6.22	55.66	5.48	59.90	20.39	56.70	11.74
Energy	51.55	3.08	36.13	-21.11	25.59	-23.91	42.06	-14.11	38.83	-14.01
LN-Entropy	53.50	6.15	64.53	23.31	71.06	20.88	71.77	27.32	65.22	19.41
Lexical Similarity	66.56	27.05	69.52	31.73	70.58	27.62	78.88	49.17	71.38	33.89
Eigenscore	64.16	22.86	75.66	42.81	76.05	30.44	77.91	45.54	73.44	35.41
HIDE	78.49	41.68	75.71	42.45	75.55	32.40	57.23	14.99	71.74	32.88
Llama-3-8B										
Perplexity	56.68	11.82	44.16	-6.38	46.96	-1.38	39.23	-5.94	46.76	-0.47
Energy	55.98	9.88	33.51	-26.30	30.93	-18.07	27.31	-39.28	36.93	-18.44
LN-Entropy	45.33	-3.66	61.24	17.13	66.40	15.30	72.40	36.05	61.34	16.21
Lexical Similarity	66.05	25.92	63.41	22.28	65.05	20.06	76.29	44.85	67.70	28.28
Eigenscore	55.49	11.51	68.38	28.24	66.40	17.54	80.04	52.15	67.58	27.36
HIDE	69.48	24.27	76.57	44.64	78.58	33.65	64.35	28.95	72.24	32.88
Llama-3-8B-Instruct										
Perplexity	58.16	10.67	46.38	-2.87	57.74	6.97	60.95	18.78	55.81	8.39
Energy	52.26	3.91	41.37	-14.63	30.19	-19.62	38.32	-17.71	40.53	-12.01
LN-Entropy	57.33	11.06	63.65	19.21	67.65	19.26	78.61	43.82	66.81	23.34
Lexical Similarity	68.51	30.04	62.46	18.42	70.47	24.73	79.98	52.49	70.35	31.42
Eigenscore	66.57	25.98	73.33	34.87	79.08	31.70	81.32	56.83	75.07	37.34
HIDE	78.16	39.28	80.77	52.52	77.45	35.56	60.25	20.98	74.15	37.08
Gemma-2-9B										
Perplexity	59.69	19.95	46.14	-4.46	29.75	-4.55	40.46	-7.66	44.01	0.82
Energy	44.45	-9.16	53.56	2.39	63.18	2.08	71.93	15.07	58.28	2.60
LN-Entropy	63.10	22.58	59.07	13.84	41.64	-2.82	38.69	-7.53	50.63	6.52
Lexical Similarity	68.93	33.21	66.84	27.57	49.29	-0.60	62.08	10.82	61.79	17.75
Eigenscore	64.41	25.28	74.27	38.40	79.23	8.06	72.24	17.68	72.54	22.36
HIDE	58.11	14.86	74.57	40.93	74.12	20.87	80.94	46.90	71.94	30.89
Gemma-2-9B-Instruct										
Perplexity	59.65	12.24	52.92	3.96	30.70	-8.91	28.99	-7.91	43.07	-0.16
Energy	59.03	12.08	81.68	51.69	78.04	19.77	92.75	34.10	77.87	29.41
LN-Entropy	62.39	16.99	68.31	24.85	44.40	-3.54	36.64	-6.75	52.94	7.89
Lexical Similarity	73.28	39.48	73.55	35.35	53.91	3.30	49.55	0.13	62.57	19.57
Eigenscore	70.96	32.03	72.42	40.43	66.72	12.17	60.28	7.88	67.59	23.13
HIDE	66.08	20.79	72.62	37.79	84.00	41.94	67.92	38.57	72.65	34.77

Table C.1: AUC-ROC and PCC values using exact match as the correctness measure for faithfulness and factuality hallucination detection across four datasets. HIDE is compared with single-pass and multi-pass baselines (rows corresponding to multi-pass are highlighted in grey; 5 outputs are generated for each multi-pass method). The best result for each metric is shown in **bold**; the best among single-pass methods is underlined.

Example from SQuAD

Prompt: College sports are also popular in southern California. The UCLA Bruins and the USC Trojans both field teams in NCAA Division I in the Pac-12 Conference, and there is a longtime rivalry between the schools.

Q: The Bruins belong to which college?

A:

Ground-truth: UCLA

Llama-3-8B Output: The University of California, Los Angeles.

Input keywords: [UCLA, rivalry, southern, sports, Pac, schools, Trojans, belong, popular, Division, Q, field, longtime, in, both, The, also, is, 12, there]

Output keywords: [University, Angeles, California, Los, The, of]

Selected Input Tokens: [UCLA, rivalry, southern, sports, Pac, schools]

Selected Output Tokens: [University, Angeles, California, Los, The, of]

HIDE Score: 0.139

Example from NQ

Prompt: Answer these questions:

Q: Who played solomon in little house on the prairie?

A:

Ground-truth: Todd Bridges

Llama-3-8B Output: The actor who played Solomon in Little House on the Prairie was Jonathan Gilbert.

Input keywords: [solomon, prairie, house, questions, Q, little, who, played, A, the, Answer, in, these, on]

Output keywords: [Gilbert, Solomon, Jonathan, actor, Prairie, Little, House, who, was, played, The, the, in, on]

Selected Input Tokens: [sol, omon, pr, airie, house, questions, Q, little, who, played, A, the, Answer, in]

Selected Output Tokens: [Gilbert, Solomon, Jonathan, actor, Prairie, Little, House, who, was, played, The, the, in, on]

HIDE Score: 0.066

Example from NQ

Prompt: Answer these questions:

Q: Who plays joker in batman the dark knight?

A:

Ground-truth: Ledger

Llama-3-8B Output: Heath Ledger

Input keywords: [joker, batman, knight, questions, plays, dark, Q, A, who, Answer, these, in, the]

Output keywords: [Ledger, Heath]

Selected Input Tokens: [joker, bat]

Selected Output Tokens: [Ledger, Heath]

HIDE Score: 0.249

Example from TriviaQA

Prompt: Answer these questions:

Q: Who was the man behind The Chipmunks?

A:

Ground-truth: David Seville

Llama-3-8B Output: Ross Bagdasarian, Sr.

Input keywords: [Chipmunks, questions, Who, Q, The, the, A, behind, was, man, Answer, these]

Output keywords: [Chipmunks, Chipmunk, Bagdasarian, album, Songbook, Ross, name, first, The, the, of, Sr, A, Q, was, What]

Selected Input Tokens: [Chip, m, unks, questions, Who, Q, The, the, A, behind, was, man, Answer,

Table D.1: Historical full-cohort AUC-ROC and PCC values under the original stopping protocol, using sentence similarity and ROUGE-L as correctness measures for faithfulness and factuality hallucination detection across four datasets. HIDE is compared with single-pass and multi-pass baselines (rows corresponding to multi-pass are highlighted in gray; five outputs are generated for each multi-pass method). The best result for each metric is shown in **bold**; the best among single-pass methods is underlined.

Method	Faithfulness						Factuality						Average					
	RACE			SQuAD			NQ			TriviaQA			AUC _s			PCC _s		
	AUC _s	AUC _c	PCC _s	AUC _s	AUC _c	PCC _s	AUC _s	AUC _c	PCC _s	AUC _s	AUC _c	PCC _s	AUC _s	AUC _c	PCC _s	AUC _s	AUC _c	PCC _s
Perplexity	50.27	49.94	18.6	15	46.53	47.3	-7.81	-1.56		Llama-3.2-3B	53.03	54.25	-8.16	9.2	57.53	57.01	6.14	20.41
Energy	49.07	49.93	13.10	9.14	36.9	37.22	-29.49	-26.22	36.83	35.2	-25.81	-10.75	43.31	41.84	-34.53	-22.20	-12.51	10.76
LN-Entropy	50.90	50.26	6.01	6.96	60.85	60.89	14.24	19.74	72.39	71.31	20.58	22.53	67.24	22.52	28.82	62.86	62.42	15.84
Lexical Similarity	64.11	65.04	22.04	27.32	64.61	66.71	28.81	34.1	65.98	69.51	20.32	27.8	79.74	81.34	52.99	56.59	70.65	31.04
Eigenscore	58.17	57.18	9.56	12.37	65.31	65.49	21.29	25.46	68.34	67.63	20.27	17.39	76.13	76.57	50.45	48.15	66.98	25.85
HIDE	66.43	65.48	18.34	20.72	77.55	77.28	49.86	50.3	75.16	74.63	47.77	25.77	58.36	59.14	27.46	17.88	69.37	35.86
Llama-3.2-3B-Instruct																		
Perplexity	59.52	61.28	34.59	28.44	52.50	55.22	9.73	15.36	54.90	54.16	-7.26	8.80	60.08	60.14	6.42	22.49	56.75	57.70
Energy	54.19	57.03	21.48	17.04	37.36	38.57	-18.23	-16.81	25.89	24.72	-48.16	-25.99	42.51	41.59	-31.54	-16.17	39.99	40.48
LN-Entropy	54.46	55.65	14.96	12.96	62.98	64.74	23.51	29.69	71.88	70.42	25.68	24.93	71.83	72.01	22.95	29.18	65.29	65.70
Lexical Similarity	66.75	68.20	31.91	33.83	69.05	71.69	35.08	41.91	70.49	70.53	34.62	32.62	79.25	80.59	51.98	54.42	71.38	72.75
Eigenscore	62.82	62.09	19.09	20.92	73.14	73.57	39.38	45.56	76.42	75.56	41.64	32.71	78.35	79.03	53.46	49.55	72.68	72.57
HIDE	74.88	73.02	27.36	33.95	75.18	75.48	42.80	44.98	76.56	77.27	53.91	35.40	58.43	59.03	30.64	19.54	71.26	71.20
Llama-3-8B																		
Perplexity	50.78	51.93	23.01	18.07	45.08	46.37	-8.01	-2.39	48.18	47.51	-11.22	2.16	38.21	36.90	-19.49	-8.54	45.56	45.68
Energy	50.10	51.35	17.68	12.31	35.27	35.48	-29.16	-26.95	33.70	32.13	-23.86	-15.02	27.43	26.48	-50.84	-42.86	36.63	36.36
LN-Entropy	45.49	45.85	-0.10	-0.54	60.50	60.57	15.09	19.82	65.81	65.00	13.80	16.17	72.15	72.08	33.54	37.79	60.99	60.87
Lexical Similarity	66.10	66.92	29.15	33.80	64.01	66.57	29.36	33.69	65.22	65.84	20.10	25.87	77.13	78.65	47.18	50.24	68.12	69.50
Eigenscore	54.16	53.98	6.73	10.88	66.57	66.34	25.66	29.93	64.36	64.76	17.07	16.90	81.06	82.14	57.92	57.46	66.54	66.81
HIDE	69.81	66.95	16.76	22.87	76.87	77.45	49.09	50.43	79.02	79.17	49.61	33.91	65.65	66.82	41.82	35.19	72.84	72.60
Llama-3-8B-Instruct																		
Perplexity	60.20	61.91	22.52	22.42	47.46	50.73	2.95	6.94	57.47	56.11	-7.64	9.72	61.70	61.14	7.68	21.43	56.71	57.47
Energy	55.94	57.16	18.26	15.97	43.05	44.60	-8.96	-8.31	30.63	29.56	-36.02	-18.50	39.40	38.80	-29.22	-16.69	42.25	42.53
LN-Entropy	57.13	57.83	12.59	14.26	62.73	63.44	23.11	26.94	67.29	65.43	18.12	20.80	79.61	79.72	42.76	48.41	66.69	66.60
Lexical Similarity	69.28	71.22	36.64	40.07	62.46	65.14	25.76	29.69	69.05	68.88	26.24	28.84	81.12	82.33	58.68	59.19	70.48	71.89
Eigenscore	63.65	62.99	18.78	23.01	72.11	71.49	36.95	39.85	77.89	77.53	41.23	32.73	82.83	83.94	69.32	63.72	74.12	73.99
HIDE	73.20	72.73	22.12	30.11	80.15	79.14	52.27	55.98	79.29	78.91	56.33	37.98	61.70	62.94	36.73	27.27	73.58	73.43
Gemma-2-9B																		
Perplexity	59.88	60.65	29.89	27.98	47.42	48.29	-8.61	-1.85	29.92	33.59	9.29	12.95	43.08	46.18	-4.69	6.65	45.07	47.68
Energy	44.05	43.20	-13.38	-12.32	52.12	52.50	6.81	2.25	64.89	60.48	0.50	-3.93	68.90	67.80	14.82	7.61	57.49	56.00
LN-Entropy	63.27	64.11	25.93	26.76	60.05	61.32	15.72	22.29	44.14	53.01	8.19	20.20	41.53	44.85	0.95	13.07	52.25	55.81
Lexical Similarity	69.02	69.62	34.18	35.48	67.10	70.32	35.25	40.56	56.80	59.43	13.50	23.50	62.98	65.46	23.79	32.99	63.97	66.21
Eigenscore	64.58	64.35	24.99	25.20	73.40	75.14	41.66	47.29	73.31	74.31	7.70	16.79	71.40	71.70	23.15	27.42	70.67	71.37
HIDE	58.06	57.99	20.60	16.50	75.17	76.66	47.63	48.43	79.08	75.96	24.08	15.41	80.84	81.02	50.51	43.88	73.29	72.91
Gemma-2-9B-Instruct																		
Perplexity	60.22	61.27	14.11	17.12	53.15	56.90	5.73	12.49	31.52	34.14	29.54	0.19	30.89	36.85	52.60	11.92	44.25	47.29
Energy	57.29	55.87	7.28	5.99	80.08	77.41	49.54	50.08	77.19	76.56	-9.51	11.55	92.34	90.96	-12.88	28.05	76.73	75.20
LN-Entropy	62.82	64.58	17.67	20.84	67.09	69.50	26.65	33.47	44.68	46.01	13.79	4.64	39.70	44.13	28.49	14.62	53.57	56.06
Lexical Similarity	72.55	72.98	34.16	38.63	72.33	74.97	39.77	46.57	53.27	56.91	19.80	11.13	51.56	56.22	31.36	21.47	62.43	65.27
Eigenscore	70.22	70.59	24.78	29.44	71.70	73.99	44.71	49.87	65.97	67.11	8.40	14.46	61.17	64.63	15.05	21.13	67.26	69.08
HIDE	64.18	64.03	11.92	16.26	72.69	73.32	39.73	39.70	85.19	85.89	51.20	41.99	70.90	70.72	52.64	45.90	73.24	73.49

these]

Selected Output Tokens: [Chip, m, unks, Chip, m, unk, Bag, das, arian, album, Song, book, Ross, name]

HIDE Score: 0.066

Example from TriviaQA

Prompt: College sports are also popular in southern California. The UCLA Bruins and the USC Trojans both field teams in NCAA Division I in the Pac-12 Conference, and there is a longtime rivalry between the schools.

Q: What is the Japanese share index called?

A:

Ground-truth: Nikkei

Llama-3-8B Output: The Nikkei

Input keywords: [index, Japanese, share, Q, questions, A, Answer, these, the, What, called, is]

Output keywords: [Nikkei, The]

Selected Input Tokens: [index, Japanese, share, Q]

Selected Output Tokens: [Nik, ke, i, The]

HIDE Score: 0.187

Appendix E: Prevalence and Impact of Very Short Outputs

HSIC-based dependence estimation requires multiple output tokens to meaningfully assess statistical coupling between input and output representations. Consequently, HIDE is not informative when the generated output consists of a single token and is less reliable for extremely short responses.

To contextualize this limitation, Table E.1 reports the prevalence of very short model-generated outputs for Llama-3-8B across QA datasets. We observe a clear dataset-dependent pattern: TriviaQA, which emphasizes factual recall, produces a substantially higher fraction of single-token and two-token answers, whereas SQuAD and RACE more often elicit short phrases or sentence-level responses.

Although extremely short outputs represent a genuine edge case for HSIC-based dependence estimation, they do not dominate the overall evaluation and primarily affect datasets with inherently minimal answer lengths. This analysis complements the discussion in Section 8.1 and explains why the short-output failure mode does not alter the main performance trends reported in Section 6.

Dataset	% length = 1	% length ≤ 2
SQuAD	14.9	28.8
RACE	11.1	16.6
NQ	12.0	31.1
TriviaQA	36.6	71.5

Table E.1: Prevalence of very short model-generated outputs for Llama-3-8B across QA datasets.

Model	Dataset	N_+	HIDE		Attention Ω		Norm $\tilde{\Delta}_{in}$	
			AUC_r	PCC_r	AUC_r	PCC_r	AUC_r	PCC_r
Llama-3-8B	SQuAD	922	77.32	48.97	85.57	60.71	51.24	-0.69
Llama-3-8B	RACE	875	58.08	9.35	59.88	14.16	47.00	-11.84
Llama-3-8B	NQ	280	78.58	31.79	84.00	32.58	71.44	22.36
Llama-3-8B	TriviaQA	1269	58.58	5.56	73.38	43.02	64.69	26.77
Gemma-2-9B	SQuAD	849	76.51	48.85	87.58	65.46	54.13	9.15
Gemma-2-9B	RACE	1185	58.12	16.29	50.41	-1.90	50.95	1.93
Gemma-2-9B	NQ	480	63.16	16.47	69.10	23.10	60.08	13.96
Gemma-2-9B	TriviaQA	1359	51.18	-5.81	66.30	32.04	62.86	21.04

Table F.1: Matched detector comparison under first-answer-line stopping: 2,000 fixed seed-42 examples per model/dataset. N_+ counts answers with ROUGE-L greater than 0.5. AUC and PCC are multiplied by 100; PCC uses the continuous reference metric. All scores use the same examples and labels; higher raw scores predict correctness. No direction is selected using test performance. These results are separate from the historical full-cohort tables.

Model	Dataset	HIDE – attention	95% CI	HIDE – norm
Llama-3-8B	SQuAD	-9.01	[-11.43, -6.69]	+22.55
Llama-3-8B	RACE	-3.00	[-5.27, -0.82]	+11.69
Llama-3-8B	NQ	-5.77	[-9.66, -2.21]	+6.42
Llama-3-8B	TriviaQA	-16.67	[-19.96, -13.56]	-7.67
Gemma-2-9B	SQuAD	-10.82	[-13.37, -8.56]	+21.58
Gemma-2-9B	RACE	+6.69	[+3.70, +9.69]	+7.37
Gemma-2-9B	NQ	-5.23	[-8.69, -1.86]	+4.11
Gemma-2-9B	TriviaQA	-17.11	[-20.84, -13.51]	-11.70

Table F.2: Paired AUC_s differences in percentage points on the corrected 2,000-example cohorts. Intervals use 2,000 paired example bootstrap draws (seed 42), are pointwise, and are not adjusted for multiple comparisons.

Appendix F: Corrected Paired Comparison: Additional Diagnostics

Across the 16,000 selected records, we retain 252 generations reaching the 256-token cap, nine empty evaluated answers, and seven no-output-state zero fallbacks (all seven in Gemma/NQ). These sets can overlap. No record is excluded from the paired metrics.

These analyses use the same fixed corrected cohorts as Table 4, without score filtering or class balancing. Table F.1 reports ROUGE-L-based metrics; Table F.2 reports paired AUC_s differences. PCC in the detector tables uses continuous reference similarity, whereas the structural plots correlate detector scores with one another. These are distinct quantities.

For realized selected count $n \geq 1$, substituting all-ones Gram matrices into the adapted score gives $c(n) = (n - 1)/n^2$ (and we define $c(0) = 0$ for the no-state fallback). Table F.3 compares this diagnostic with HIDE on exactly the same examples. The near-unit correlations and small AUC differences indicate that the configured score is dominated by selected-token count in these runs; they do not establish a nonlinear-dependence advantage. We did not retune the kernel on the test labels.

Model	Dataset	HIDE AUC _s	Count-control AUC _s	PCC(HIDE, control)
Llama-3-8B	SQuAD	76.33	76.38	1.000000
Llama-3-8B	RACE	61.66	61.92	1.000000
Llama-3-8B	NQ	78.47	78.57	1.000000
Llama-3-8B	TriviaQA	57.01	56.58	1.000000
Gemma-2-9B	SQuAD	75.90	75.92	0.998759
Gemma-2-9B	RACE	57.93	57.83	0.998718
Gemma-2-9B	NQ	62.74	63.01	0.999911
Gemma-2-9B	TriviaQA	49.83	49.78	0.999988

Table F.3: Selected-token-count diagnostic on the same corrected cohorts. The control is $(n_{\text{eff}} - 1)/n_{\text{eff}}^2$ for $n_{\text{eff}} \geq 1$, and zero otherwise. AUC is multiplied by 100; PCC is unscaled. This is a control for realized selected-token count, not raw output length.

Model	Dataset	Mean input tokens	Mean generated tokens
Llama-3.2-3B	SQuAD	186.47	33.44
Llama-3.2-3B	NQ	20.30	17.34
Llama-3-8B	SQuAD	186.47	21.39
Llama-3-8B	NQ	20.30	20.50
Gemma-2-9B	SQuAD	186.20	39.29
Gemma-2-9B	NQ	21.14	22.16
Gemma-2-27B	SQuAD	186.20	45.65
Gemma-2-27B	NQ	21.14	12.14

Table F.4: Mean token lengths for the timing queries in Table 6, under the earlier stopping protocol (maximum 256 generated tokens). Means include all queries and measured repeats; repeats share deterministic token sequences. These length differences limit a parameter-only interpretation of the scaling comparison.

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